Intermediate steps of Lowering in Tigrinya*

Coppe van Urk, QMUL

Abstract

This paper presents a pattern of complementizer displacement in the Ethio-Semitic language Tigrinya (Ethiopia, Eritrea) that argues for the existence of a morphosyntactic operation of lowering that may involve successive steps (Embick and Noyer 2001; Arregi and Pietraszko 2021). In Tigrinya, underlyingly clause-final complementizers may displace leftward and downward throughout a head-final verb cluster. This displacement is associated with morphological consequences, in the form of stem allomorphy and the spreading of relative morphology. Importantly, these effects show up also in intermediate positions, on auxiliaries that are crossed by lowering. As a result, the morphosyntax of Tigrinya complementizers provides evidence for intermediate steps of lowering, and against proposals in which lowering effects do not involve the dislocation of a morpheme (e.g. Brody 2000; Adger et al. 2009; Platzack 2013; Svenonius 2016).

Keywords: Tigrinya, morphology, syntax, movement, lowering, complementizers

1 Introduction

Much work on morphology and syntax recognizes the need for operations of word formation that take a morpheme associated with a specific syntactic position and combine it with a base in a lower position (Marantz 1984; Halpern 1995; Brody 2000; Embick and Noyer 2001; Adger et al. 2009; Harley 2013; Harizanov and Gribanova 2019; Arregi and Pietraszko 2021). But different conceptions of lowering exist in the literature, with disagreement about what and how many operations are needed, whether lowering effects can be phonological, morphological or syntactic in nature, and to what extent such mechanisms are similar to those that affect upward movement in syntax.

This paper presents a pattern of complementizer displacement in the Ethio-Semitic language Tigrinya (Ethiopia, Eritrea) that offers insight into the underlying mechanisms that achieve lowering effects. In Tigrinya, various complementizers, such as the declarative complementizer *käm* and the question particle *do* (in **bold**) move leftward and apparently downward, onto auxiliaries and verbs in a head-final cluster, as shown in (1a–b) and (2a–c).

(1) *The complementizer käm undergoes leftward displacement:*

a. ካብ ገዛ ትወጽእ ከምዝነበረት ሓቢረ።
[CP kab gäza tɨ-wäts'ɨʔ **käm**=zɨ-näbär-ät] ħabir-ä
from house 3FS-leave.IMPF C=REL-AUX.PFV.PST-3FS report-1S
'I reported that she was leaving the house.'

b. ካብ ገዛ ከምትወጽእ ዝነበረት ሓቢረ።
[CP kab gäza **käm**=tɨ-wäts'ɨʔ zɨ-näbär-ät] ħabir-ä
from house C=3FS-leave.IMPF REL-AUX.PFV.PST-3FS report-1S
'I reported that she was leaving the house.'

(2) *The question particle do displaces leftward:*

a. እቲ ቆልዓ ደቂሱ ኔሩ ዶ፧
ʔɨt-i qolʕa däk'is-u ner-u **do**?
DEF-MS baby sleep.GER-3MS AUX.PST-3MS Q

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‘Had the baby slept?’

b. **እቲ ቆልዓ ደቂሱ ዶ ኔሩ፧**
ʔɨt-i qolʕa däk’is-u **do** ner-u?
DEF-MS baby sleep.GER-3MS Q AUX.PST-3MS
‘Had the baby slept?’

c. **እቲ ቆልዓ ዶ ደቂሱ ኔሩ፧**
ʔɨt-i qolʕa **do** däk’is-u ner-u?
DEF-MS baby Q sleep.GER-3MS AUX.PST-3MS
‘Had the baby slept?’

This paper proposes that complementizer placement in Tigrinya reflects a postsyntactic operation of Lowering (Embick and Noyer 2001; Harizanov and Gribanova 2019), which adjoins a head-final complementizer such as *käm* onto a lower functional head, like the past tense auxiliary *näbär-* (3), as in the example in (1a).

(3) *Lowering onto the auxiliary:*
[$_V$ tɨ-wäts’ɨʔ$_V$] [$_T$ **käm**= [$_T$ zɨ-näbär-ät$_{Aux}$]] käm=

Key motivation for the idea of lowering comes from the observation that complementizer displacement can be *non-local*, skipping over an intervening auxiliary, as in (1b) and (2b). I argue that such examples reflect successive applications of Lowering (4), involving a step of excorporation (e.g. Roberts 1991; Güneş 2021).

(4) *Successive Lowering onto the verb:*
[$_V$ **käm**= [$_V$ tɨ-wäts’ɨʔ$_V$] [$_T$ käm= [$_T$ zɨ-näbär-ät$_{Aux}$]] käm=

I develop a conception of Lowering that allows such derivations. First, following Adamson (2022) and Weisser (2024, 2025), I propose that Lowering targets the closest head, and not necessarily the head of its complement (see also Embick and Noyer 2001:sec. 7.2). Second, I propose that excorporation is permitted in postsyntactic operations (Güneş 2021; cf. Roberts 1991). Under a definition of c-command in which a head adjoined to a complex head c-commands out (Kayne 1994:ch. 3.2), these assumptions allow for structures like (4) with successive Lowering. Independent support for such derivations comes from non-local lowering in Tiwa (Dawson 2017) and varieties of German (Schallert 2020).

This account makes use of an operation of Lowering that takes place before Vocabulary Insertion and one in which non-local Lowering reflects *successive applications* of the same operation (Embick and Noyer 2001; Arregi and Pietraszko 2021). Tigrinya provides important evidence for both of these characteristics of lowering, because complementizer displacement is associated with two morphological effects: i) Lowering triggers the insertion of the relative prefix *zɨ-* on any verb/auxiliary without a prefix (in *italics*), as in (5a–b) and (6a–b), and ii) this relative prefix triggers the perfective stem allomorph of verbs/auxiliaries and the associated agreement suffixes (<u>underlined</u>), as evident in (6a–b).

(5) *Insertion of zɨ- prefix on auxiliary and verb:*

a. **እቲ ህጻን ይድቅስ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።**
[$_{CP}$ ʔɨt-i häts’an jɨ-dɨk’ɨs **sɨlä** *z*-äll-o] ʔɨt-i sɨbʔaj ħagus ʔɨjj-u.
DEF-MS baby 3MS-sleep.IMPF because REL-AUX-3MS DEF-MS man happy COP-3MS
‘Because the baby is sleeping, the man is happy.’

b. **እቲ ህጻን ስለ ዝድቅስ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።**
[$_{CP}$ ʔɨt-i häts’an **sɨlä** *zɨ*-dɨk’ɨs *z*-äll-o] ʔɨt-i sɨbʔaj ħagus ʔɨjj-u.
DEF-MS baby because REL-sleep.IMPF REL-AUX-3MS DEF-MS man happy COP-3MS
‘Because the baby is sleeping, the man is happy.’

(6) *The prefix zɨ- triggers stem and suffix allomorphy:*

a.

	ካብ	ገዛ	ትወጽእ	ከምዝነበረት	ሓቢረ።
[CP	kab	gäza	tɨ-wäts'ɨʔ	**käm**=*zɨ*-näbär-ät]	ħabir-ä
	from	house	3FS-leave.IMPF	C=REL-AUX.PFV.PST-3FS	report-1S

'I reported that she was leaving the house.'

b.

	ካብ	ገዛ	ከምትወጽእ	ዝነበረት	ሓቢረ።
[CP	kab	gäza	**käm**=tɨ-wäts'ɨʔ	*zɨ*-näbär-ät]	ħabir-ä
	from	house	C=3FS-leave.IMPF	REL-AUX.PFV.PST-3FS	report-1S

'I reported that she was leaving the house.'

Importantly, these morphological effects still occur in *intermediate position*, so that the auxiliaries in (5b) and (6b) show the reflexes of Lowering, even though the complementizer undergoes a further step of Lowering onto the verb. As a result, Tigrinya complementizer placement offers novel evidence that successive lowering involves intermediate steps (Embick and Noyer 2001; Arregi and Pietraszko 2021). These facts argue against a conception of lowering in which it occurs in "one fell swoop", as, for example, in Harizanov and Gribanova's (2019) mechanism of amalgamation, or one in which lowering effects are not the result of a process of dislocation, such as Brody's (2000) Mirror Theory and its descendents (Adger et al. 2009; Svenonius 2016), in which apparent lowering reflects the parametrization of realization within an extended projection (see also Platzack 2013).

The paper is structured as follows. Section 2 introduces the morphosyntax of auxiliaries and verbs, showing that Tigrinya is an SOV language with an opposition between perfective and imperfective stems. In section 3, I show that several complementizers do not appear clause-finally, but instead displace leftward throughout the head-final verbal cluster. Section 4 argues for an analysis of complementizer placement in terms of Embick and Noyer's (2001) postsyntactic operation of Lowering. In section 5, I show how such an approach can be extended to explain instances of successive Lowering, as evidenced by morphological reflexes of Lowering in intermediate position.

2 Verbs and auxiliaries in Tigrinya

This section introduces the morphology of syntax of verbs and auxiliaries in Tigrinya. Tigrinya is generally SOV, with an opposition between perfective and imperfective verb stems, distinguished by gemination, vocalic melody, and agreement affixes. These stems can combine with a number of different head-final auxiliaries to express different tense/aspect combinations. Although the verb and auxiliary form a cluster, they behave as independent words, each with its own vocalic melody and set of affixes.

2.1 Imperfective and perfective verb stems in Tigrinya

I first provide an overview of the morphosyntax of verbal morphology in Tigrinya, which will be important to understand exactly how this system is affected by proclitic complementizers. Tigrinya is SOV, and the imperfective and gerund stem can be used in declaratives without any other TAM marking, expressing the present imperfective and past perfective, respectively. These stems are distinguished by gemination, vocalic melody and agreement affixes. In addition, Tigrinya displays an allomorphy pattern with perfective stems, in the form of an alternation between the gerund form and traditional perfective.

Tigrinya is an Ethio-Semitic language spoken in Eritrea and Northern Ethiopia, with an estimated 10 million speakers. The data presented here comes from three speakers originally from Asmara. It was collected in two field methods classes as well as elicitation sessions in 2021, 2023, 2024, and 2025.

Tigrinya is a head-final SOV language. In a neutral declarative clause, the verb must appear clause-finally (7a–c) (though see Gebreghziaber and Duguine 2024 on word order in *wh*-questions).

(7) *Tigrinya is SOV:*

a. **እቲ ከልቢ ቡን ይሰቲ።**
ʔɨt-i kälbi bun **jɨ-säti**.
def-ms dog coffee 3ms-drink.impf
'The dog drinks coffee.'

b. ***እቲ ከልቢ ይሰቲ ቡን።**
*ʔɨt-i kälbi **jɨ-säti** bun.
def-ms dog 3ms-drink.impf coffee
'The dog drinks coffee.'

c. ***ከልቢ እቲ ይሰቲ ቡን።**
***jɨ-säti** ʔɨt-i kälbi bun.
3ms-drink.impf def-ms dog coffee
'The dog drinks coffee.'

Finite verbs take at least three forms (Leslau 1941:ch. 5; Kifle 2011). As in many Semitic languages, verbal morphology is built around an opposition between the perfective and imperfective verb stem, both of which are used in a range of TAM combinations (Meyer 2016).[1] The imperfective stem in isolation typically expresses present imperfective meanings (7a). The perfective stem has two allomorphs. An innovation of Tigrinya is that the gerund stem has taken over some of the functions of the historical perfective stem. The gerund allomorph of the perfective stem is used in isolation to express past perfective meanings (8a–c). This allomorphy pattern interacts with complementizer lowering and will play an important role in diagnosing intermediate steps of lowering in section 5.

(8) *Gerund allomorph in plain declarative:*

a. **እቲ ከልቢ ነቲ ስጋ በሊዕዎ።**
ʔɨt-i kälbi n-ät-i sɨga **bäliʕ-ɨw-o**.
def-ms dog acc-def-ms meat eat.ger-3ms-3ms.o
'The dog ate the meat.'

b. **ጸጋ ትማሊ ደሪፋ።**
Ts'äga tɨmali **därif-a**.
Ts'äga yesterday sing.ger-3fs
'Ts'äga sang yesterday.'

c. **ቱፋሕ ገዚአ።**
tufaħ **gäziʔ-ä**.
apple buy.ger-1s
'I bought apples.'

The historical perfective stem still appears in negated clauses, after the negative clitic *ʔaj*= (9a–b), as well as after relative morphology (Leslau 1941:83), presumably reflecting the origins of the gerund stem in reduced clauses.

(9) *Perfective allomorph after negative clitic ʔaj=:*

a. **እቲ ከልቢ ወላሓንቲ ኣይበለዐን።**
ʔɨt-i kälbi walla-ħanti ʔaj=**bäläʕ-ä**-n.
def-ms dog even-one neg=eat.perf-3ms-neg
'The dog didn't eat anything.'

[1] I will use these labels throughout, but it is important to note that these stem types then are not necessarily always associated purely with perfective and imperfective meanings. The perfective is also used to express the perfect, for example, while only the imperfective appears in the future tense.

b. **ጻጋ ትማሊ ኣይደረፈትን።**
Ts’äga tɨmali ʔaj=**däräf-ät**-ɨn.
Ts’äga yesterday NEG=sing.PERF-3FS-NEG
‘Ts’äga sang yesterday.’

I will use the term *perfective stems* to refer to the underlying syntax that surfaces with this allomorphy pattern, and I refer to the allomorphs as the *gerund allomorph* and *perfective allomorph*, respectively.

Table 1 shows the morphophonological differences between these three different stems for different verb classes.[2]

Table 1. Tigrinya verb stems (Kifle 2011:41).

Root	Gerund	Perfective	Imperfective
(A) s-b-r ‘break’	säbir-u	säbär-ä	jɨ-säbbɨr
(B) f-ʃ-m ‘finish’	fäʃʃim-u	fäʃʃäm-ä	jɨ-fɨʃʃɨm
(C) b-r-k’ ‘bless’	barik’-u	baräk’-ä	jɨ-barɨk’

Roots are triconsonantal units, and different verb stems are formed by imposing different vocalic melodies and sometimes gemination of the medial consonant, with variation by verb class.

As in other Semitic languages, the imperfective and perfective/gerund stems differ in how they are marked by agreement affixes (Leslau 1941; Kifle 2011). The perfective stems combine with agreement suffixes, while the imperfective stem additionally makes use of prefixes. Table 2 provides an overview.

Table 2. Agreement affixes for the verb *s-b-r* ‘break’ (Kifle 2011:42).

	Gerund	Perfective	Imperfective
1SG	säbir-ä	säbär-ku	**jɨ**-säbbɨr[3]
1PL	säbir-na	säbär-na	**nɨ**-säbbɨr
2FS	säbir-ki	säbär-ki	**tɨ**-säbɨr-i
2MS	säbir-ka	säbär-ka	**tɨ**-säbbɨr
2FP	säbir-kɨn	säbär-kɨn	**tɨ**-säbɨr-a
2MP	säbir-kum	säbär-kum	**tɨ**-säbɨr-u
3FS	säbir-a	säbär-ät	**tɨ**-säbbɨr
3MS	säbir-u	säbär-ä	**jɨ**-säbbɨr
3FP	säbir-än	säbär-a	**jɨ**-säbɨr-a
3MP	säbir-om	säbär-u	**jɨ**-säbɨr-u

As evident in Table 2, the perfective stems share many agreement suffixes, but combine with different suffixes across the 3rd person and in the 1st person singular. These differences are not obviously phonologically derivable from the different vocalic melodies, and so this allomorphy pattern persists in suffixes also.

I adopt a head-final syntax for Tigrinya, with the verb raising to Asp and then to T. An example like (10a) then has a structure like (10b).

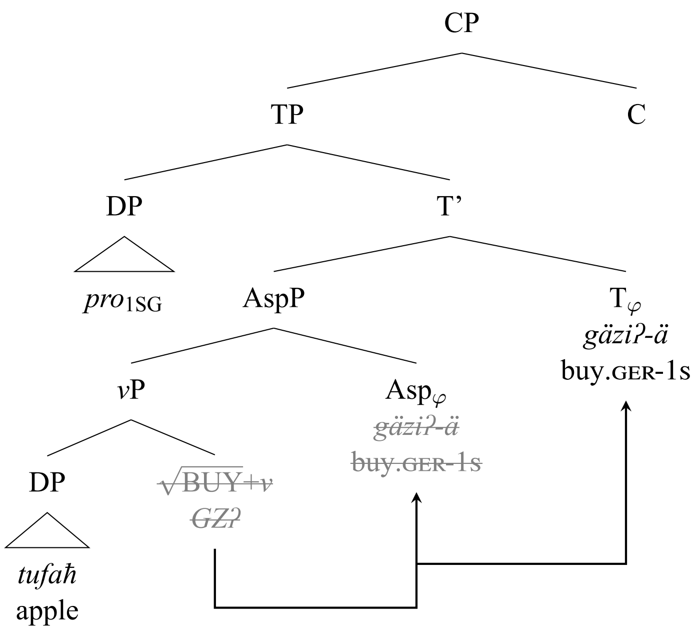
(10) *A head-final syntax for Tigrinya:*

a. **ቱፋሕ ገዚአ።**
tufaħ **gäziʔ-ä**.
apple buy.GER-1s
‘I bought apples.’

[2]I will set aside the jussive, which appears in main clauses and so does not interact with the complementizer system that is the focus of this paper.

[3]Bulakh (2019:fn. 7) notes that some speakers may realize the 1SG prefix as *ʔɨ-*.

b.



I posit φ-probes on Asp and T, both of which target the subject. On the verb, subject agreement is realized only once, which I attribute to a constraint prohibiting multiple agreement morphemes expressing the same feature on one host (Kinyalolo 1991). As a result, when the verb moves through Asp and to T, only one agreement suffix appears. As we will see, both instances of subject agreement are expressed in verb+auxiliary combinations, when separate hosts are available.

To capture the stem alternations, I propose that the vocalic melodies indicating perfective and imperfective are realizations of Asp, reflecting Vocabulary Insertion rules like those in (11) for root class A.[4] The imperfective melody spells out progressive and habitual flavors of Asp, while the perfective stems realize perfective or perfect aspect.[5] I treat the gerund stem allomorph as the elsewhere form, while the perfective stem is conditioned by the presence of Neg or relative morphology.[6]

(11) *Insertion rules for aspectual melodies (Root class A):*
Asp$_{\text{PROG/HAB}}$ $\rightarrow$ *ä ɨ*
Asp$_{\text{PFV/PERF}}$ $\rightarrow$ *ä i*
Asp$_{\text{PFV/PERF}}$ $\rightarrow$ *ä ä* / Neg/Rel ___

We have seen then that Tigrinya verbs appear clause-finally and show a morphological opposition between imperfective and perfective aspect. In the next section, we will see that these verb stems combine with head-final auxiliaries to express different tense/aspect combinations.

2.2 The morphosyntax of Tigrinya auxiliaries

Tigrinya verb stems may be combined with a single auxiliary to express a range of different tense/aspect combinations (Leslau 1941; Meyer 2016). The present auxiliary *ʔall-* and past auxiliary *ner-* combine with the perfective stems to express the perfect and with the imperfective to form the progressive. In

[4]These rules treat the first vowel *ä* as part of the aspectual melody, though it is constant across all three aspect allomorphs for Root Class A. The first vowel appears to be a marker of verb class, perhaps an exponent of *v*, since only this vowel varies across root classes. In addition, Root Class C appears to be distinguished from the other classes only by the use of *a* as first vowel. See Table 1.

[5]For ease of exposition, I adopt a clause structure with only one Asp head, but some of these aspects may well involve additional functional structure.

[6]There are a few options for how to ensure that the gerund/perfective allomorphy pattern extends into the agreement suffixes. One is to treat Neg and Rel as the trigger for this allomorphy pattern, and posit non-local allomorphy (Božič 2019). Another option would be to distinguish the gerund and perfective allomorphy by means of a feature. In that view, Neg and Rel could be associated with a rule of feature insertion in the sense of Noyer (1998).

addition, the stative auxiliary *ʔijj-* marks the future in combination with the imperfective. All of these auxiliaries appear head-finally. In addition, we will see that auxiliaries display the same allomorphy pattern as perfective stems.

A number of TAM combinations involve a verb and a single auxiliary. These auxiliaries appear clause-finally also, in main and embedded clauses (12a–c).

(12) *Auxiliaries are head-final and appear after the verb:*

a. እቲ ከልቢ ነቲ ስጋ ይበልዖ **ኣሎ**።
ʔɨt-i kälbi n-ät-i sɨga ji-bälʕ-o **ʔall-o**.
DEF-MS dog ACC-DEF-MS meat 3MS-eat.IMPF-3MS.O AUX-3MS
'The dog is eating the meat.'

b. ጸጋ ትደርፍ **ኔራ**።
Ts'äga tɨ-därrɨf **ner-a**.
Ts'äga 3FS-sing.IMPF AUX.PST-3FS
'Ts'äga was singing.'

c. ካብ ገዛ ወጺኣ **ከምዘላ** ሓቢረ።
[CP kab gäza wäts'iʔ-a **käm=z-äll-a**] ħabir-ä
from house leave.GER-3FS C=REL-AUX-3FS report.GER-1S
'I reported that she has left the house.'

Placing the auxiliary before the verb or an object is ungrammatical (13a–c).

(13) *Auxiliaries cannot appear preverbally:*

a. *እቲ ከልቢ ነቲ ስጋ **ኣሎ** ይበልዖ።
*ʔɨt-i kälbi n-ät-i sɨga **ʔall-o** jɨ-bälʕ-o.
DEF-MS dog ACC-DEF-MS meat AUX-3MS 3MS-eat.IMPF-3MS.O
'The dog is eating the meat.'

b. *እቲ ከልቢ **ኣሎ** ነቲ ስጋ ይበልዖ።
*ʔɨt-i kälbi **ʔall-o** n-ät-i sɨga jɨ-bälʕ-o.
DEF-MS dog AUX-3MS ACC-DEF-MS meat 3MS-eat.IMPF-3MS.O
'The dog is eating the meat.'

c. *ጸጋ **ኔራ** ትደርፍ።
*Ts'äga **ner-a** tɨ-därrɨf.
Ts'äga AUX.PST-3FS 3FS-sing.IMPF
'Ts'äga was singing.'

The verb and auxiliary must be adjacent, a requirement that is reminiscent of head-final verb clusters in some Germanic languages. Adverbs, for example, cannot appear in between verb and auxiliary (14a–d).

(14) *Adverbs cannot intervene between verb and auxiliary:*

a. ጸጋ ነቲ መጽሓፍ **ቀልጢፉ** ኣንቢባቶ ኔራ።
Ts'äga n-ät-i mäts'ħaf **k'ält'ifa** ʔanbib-at-o ner-a
Ts'äga ACC-DEF-3MS book quickly read.GER-3FS-3MS.O AUX.PST-3FS
'Ts'äga had read the book quickly.'

b. *ጸጋ ነቲ መጽሓፍ ኣንቢባቶ **ቀልጢፉ** ኔራ።
*Ts'äga n-ät-i mäts'ħaf ʔanbib-at-o **k'ält'ifa** ner-a
Ts'äga ACC-DEF-3MS book read.GER-3FS-3MS.O quickly AUX.PST-3FS
'Ts'äga had read the book quickly.'

c. ዮሰፍ ነቲ መዝሙር **ጸቡቕ** ዘሚርዎ ኔሩ።
Josäf n-ät-i mäzmɨr **ts'äbuq** zämir-uw-o ner-u
Josäf ACC-DEF-3MS song well sing.GER-3MS-3MS.O AUX.PST-3MS
'Josäf had sang the song well.'

d. ***ዮሰፍ ነቲ መዝሙር ዘሚርዎ ጸቡቕ ኔሩ።**
*Josäf n-ät-i mäzmɨr zämir-uw-o **ts’äbuq** ner-u
Josäf ACC-DEF-3MS song sing.GER-3MS-3MS.O well AUX.PST-3MS
‘Josäf had sang the song well.’

The primary function of auxiliaries is to mark tense in perfect and progressive constructions. The present auxiliary *ʔall-* appears alongside the perfective stems to indicate the present perfect (15a), and yields the present progressive when combined with the imperfective stem (15b).

(15) *Auxiliary ʔall- forms present perfect and present progressive:*

a. **ጸጋ ኣይደረፈትን ኣላ።**
Ts’äga ʔaj=däräf-ät-ɨn ʔall-a.
Ts’äga NEG=sing.PERF-3FS-NEG AUX-3FS
‘Ts’äga has not sung.’

b. **እቶም ቆልዑ ምስቲ ከልቢ ይጻወቱ ኣለዉ።**
ʔɨt-om k’olɨʕu mɨs-ti kälbi jɨ-ts’awɨt-u ʔall-ɨwu.
DEF-MP child.PL with-DEF.MS dog 3MP-play.IMPF-3MP AUX-3MP
‘The children are playing with the dog.’

The verb stems also combine with the past tense auxiliary *ner-* to form a past perfect or past progressive (16a–b):

(16) *Past auxiliary ner- forms past perfect or past progressive:*

a. **ጸጋ ደሪፋ ኔራ።**
Ts’äga därif-a **ner-a**.
Ts’äga sing.GER-3FS AUX.PST-3FS
‘Ts’äga had sung.’

b. **ጸጋ ትደርፍ ኔራ።**
Ts’äga tɨ-därrɨf **ner-a**.
Ts’äga 3FS-sing.IMPF AUX.PST-3FS
‘Ts’äga was singing.’

Finally, the imperfective stem with the stative copula *ʔɨjj-* indicates a future meaning. In this context, the imperfective is prefixed with the purposive *kɨ-*:

(17) *Future tense expressed by combining auxiliary with stative copula:*

ንሳቶም ቡን ክሰትዩ እዮም።
ʔɨssatom bun kɨ-sätj-u ʔɨjj-om
3MP coffee PURP-drink.IMPF-3MP COP-3MP
‘They will drink coffee.’

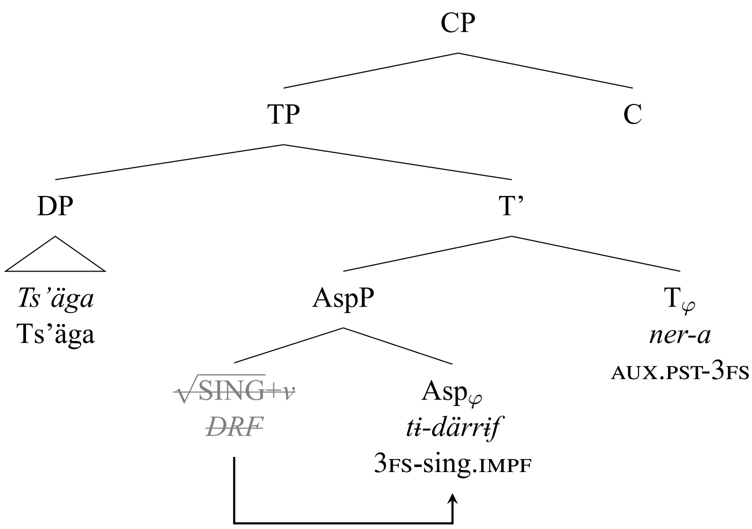
Notice that the agreement affixes on the imperfective/perfective stem do not change with the addition of an auxiliary. All auxiliaries carry an additional subject agreement suffix, yielding multiple agreement.

These observations fit with the head-final syntax I proposed in the previous section. I propose that, in the perfect and progressive, the verb remains in Asp and does not undergo further head movement to T. In these aspects then, an auxiliary provides the realization of Tense (see also Tesfay 2016). The structure in (18b) demonstrates for the example in (18a).

(18) *A head-final syntax for Tigrinya auxiliaries:*

a. **ጸጋ ትደርፍ ኔራ።**
Ts’äga tɨ-därrɨf ner-a.
Ts’äga 3FS-sing.IMPF AUX.PST-3FS
‘Ts’äga was singing.’

b.



This structure accommodates the strict head-final order observed in verbs and auxiliaries. Recall that I proposed in section 2.1 that both Asp and T carry subject agreement probes, but that multiple agreement cannot be spelled out on a single verb. In constructions with auxiliaries, the features on both probes are realized, so that both the verb and auxiliary surface with subject agreement.

Auxiliaries are independent morphological words, with a vocalic melody and separate prefixes and suffixes. Morphologically, auxiliaries have the form of the perfective stems (Kifle 2011:sec. 2.4.3). Subject agreement is expressed using the same set of agreement suffixes that are utilized for the perfective stem allomorphs. In addition, auxiliaries participate in the allomorphy pattern triggered by negation and relative morphology. The examples in (19a–b) demonstrate in the past progressive, where the choice of host for the proclitic negation *ʔaj*= (in *italics*) seems to be free.[7] When the auxiliary hosts the negative clitic, the auxiliary must surface as its perfective allomorph with the associated agreement suffixes (19b).

(19) Gerund/perfective on the past tense auxiliary:

a. **እታ ጓል ቡን ኣይትሰትን ኔሩ።**
ʔɨt-a gwal bun *ʔaj*=tɨ-sätɨ-n **ner-a**.
DEF-FS girl coffee NEG=3FS-drink.IMPF-NEG AUX.PST-3FS
’The girl was not drinking coffee.’

b. **እታ ጓል ቡን ትሰቲ ኣይነበረትን።**
ʔɨt-a gwal bun tɨ-säti *ʔaj*=**näbär-ät**-ɨn.
DEF-FS girl coffee 3FS-drink.IMPF NEG=AUX.PFV.PST-3FS-NEG
’The girl was not drinking coffee.’

The same alternations occur with the present tense auxiliaries *ʔall-* and *ʔɨjj-*. Auxiliaries then show the same alternation between a gerund and perfective allomorph, with more irregularity in the verb form, but with the expected changes in the agreement suffixes. These facts will become important when we examine the morphological effects of complementizer displacement in sections 4 and 5.[8]

With this understanding of the morphosyntax of verbs and auxiliaries in place, we can turn to the key patterns in this paper, the placement of proclitic complementizers within a head-final cluster.

3 Complementizer placement as procliticization in Tigrinya

This section introduces complementizer placement in Tigrinya and shows that a number of Tigrinya complementizers do not appear in the expected head-final position, but instead appear as proclitics in the

[7]Negation in Tigrinya is associated with two morphemes. In matrix clauses, the negative clitic is accompanied by the negative suffix *-in*.

[8]The fact that auxiliaries carry a perfective melody suggests that this melody functions as a kind of default for verbal heads.

verb+auxiliary cluster (20a–b).

(20) *Tigrinya complementizers appear as proclitics:*

a. ዮሰፍ ጸጋ ትማሊ ከምዝደረፈት ሓቢሩ።
Josäf [$_{CP}$ Ts’äga tɨmali **käm**=zɨ-däräf-ät] ħabir-u.
Josef Ts’aiga yesterday C=REL-sing.PERF-3FS say.GER-3MS
‘Josef said that Ts’aiga sang yesterday.’

b. እቲ ህጻን ደቂሱ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።
[$_{CP}$ ʔɨt-i häts’an däk’is-u **sɨlä** z-äll-o] ʔɨt-i säbʔaj ħagus ʔɨjj-u.
DEF-MS baby sleep.GER-3MS because REL-AUX-3MS DEF-MS man happy COP-3MS
‘Because the baby has slept, the man is happy.’

I argue that these Tigrinya complementizers are nonetheless underlyingly head-final, but undergo a process of leftward and downward displacement to cliticize onto a verb or auxiliary. Evidence for this conclusion comes from a number of observations. First, some complementizers do surface in the expected clause-final position. In addition, the placement of proclitic complementizers is oriented to the right edge of the clause, because they preferentially target the final verb/auxiliary. Finally, cases in which the complementizer undergoes non-local cliticization, to attach to a verb over an intervening auxiliary, demonstrate that it is the complementizer displacing leftward that gives rise to these alternations.

3.1 Proclitic complementizers in Tigrinya

A number of Tigrinya complementizers can appear as proclitics in the verb+auxiliary cluster, including at least the question particle *do*, the declarative complementizer *käm*, the adverbial complementizers *sɨlä* ‘because’ and *mɨs* ‘when’, and the interrogative complementizer *ʔɨntä*, typically attaching before the final verb or auxiliary (21a–e).[9]

(21) *Complementizers appear before final verb or auxiliary:*

a. እቲ ቆልዓ ደቂሱ ዶ ኔሩ፧
ʔɨt-i qolʕa däk’is-u **do** ner-u?
DEF-MS baby sleep.GER-3MS Q AUX.PST-3MS
‘Had the baby slept?’

b. ዮሰፍ ጸጋ ትማሊ ከምዝደረፈት ሓቢሩ።
Josäf [$_{CP}$ Ts’äga tɨmali **käm**=zɨ-däräf-ät] ħabir-u.
Josef Ts’aiga yesterday C=REL-sing.PERF-3FS say.GER-3MS
‘Josef said that Ts’aiga sang yesterday.’

c. እቲ ህጻን ደቂሱ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።
[$_{CP}$ ʔɨt-i häts’an däk’is-u **sɨlä** z-äll-o] ʔɨt-i säbʔaj ħagus ʔɨjj-u.
DEF-MS baby sleep.GER-3MS because REL-AUX-3MS DEF-MS man happy COP-3MS
‘Because the baby has slept, the man is happy.’

d. እቲ ህጻን ምስ ደቀሰ እቲ ሰብኣይ ተሓጒሱ።
[$_{CP}$ ʔɨt-i häts’an **mɨs** däk’äs-ä] ʔɨt-i säbʔaj tɨ-ħagwis-u
DEF-MS baby when sleep.PERF-3MS DEF-MS man NACT-happy.GER-3MS
‘When the baby slept, the man became happy.’

e. እቲ ሰብኣይ ነታ ሰበይቲ እቲ ቆልዓ ይድቅስ እንተ ኔሩ
ʔɨt-i säbʔaj n-ät-a säbäjti [$_{CP}$ ʔɨt-i qolʕa jɨ-dɨk’is **ʔɨntä** ner-u]
DEF-MS man DAT-DEF-FS woman DEF-MS child 3MS-sleep.IMPF whether AUX.PST-3MS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O

[9] There seems to be some variation in whether complementizers are written as separate words. I write all complementizers as separate words, except *käm*.

‘The man asked the woman whether a child was sleeping.’

All of the complementizers that mark embedded clauses are ungrammatical in clause-final position, as (22a–d) demonstrate, and so must surface as proclitics.

(22) *Complementizers ungrammatical when clause-final:*

a. *ዮሰፍ ጸጋ ትማሊ ደሪፋ ከም ሓቢሩ።
*Josäf [CP Ts’äga timali därif-a **käm**] ħabir-u.
Josef Ts’aiga yesterday sing.GER-3FS say.GER-3MS
‘Josef said that Ts’aiga sang yesterday.’

b. * እቲ ህጻን ደቂሱ ኣሎ ስለ እቲ ሰብኣይ ሕጉስ እዩ።
*[CP ʔɨt-i häts’an däk’is-u ʔall-o **sɨlä**] ʔɨt-i säbʔaj ħagus ʔɨjj-u.
DEF-MS baby sleep.GER-3MS REL-AUX-3MS because DEF-MS man happy COP-3MS
‘Because the baby has slept, the man is happy.’

c. * እቲ ህጻን ደቀሰ ምስ እቲ ሰብኣይ ትሓጐሱ።
*[CP ʔɨt-i häts’an däk’äs-ä **mɨs**] ʔɨt-i säbʔaj tɨ-ħagwis-u
DEF-MS baby sleep.GER-3MS when DEF-MS man NACT-happy.GER-3MS
‘When the baby slept, the man became happy.’

d. *እቲ ሰብኣይ ነታ ሰበይቲ እቲ ቆልዓ ይድቅስ ኔሩ እንተ]
*ʔɨt-i säbʔaj n-ät-a säbäjti [CP ʔɨt-i qolʕa jɨ-dɨk’ɨs ner-u **ʔɨntä**]
DEF-MS man DAT-DEF-FS woman DEF-MS child 3MS-sleep.IMPF whether AUX.PST-3MS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
‘The man asked the woman whether a child was sleeping.’

A number of these complementizers are prepositional in origin and are still used as prepositions also, such as *käm* ‘like’ and *mɨs* ‘with’ (23a–b).[10]

(23) *Some Tigrinya complementizers are also used as prepositions:*

a. እቶም ቆልዑ ምስቲ ከልቢ ይጻወቱ ኣለዉ።
ʔɨt-om k’olɨʕu **mɨs**-ti kälbi jɨ-ts’awɨt-u ʔall-ɨwu.
DEF-MP child.PL with-DEF.MS dog 3MP-play.IMPF-3MP AUX-3MP
‘The children are playing with the dog.’

b. ከም ገዛኺ ዝመስል ገዛ ርእየ።
käm gäza-xi zɨ-mässɨl gäza rɨʔj-ä
like house-2SG REL.3MS-look.like.IMPF house see.GER-3MS
‘I saw a house that looks like your house.’

The prepositional nature of these complementizers provides insight into how the pattern of complementizer procliticization may have arisen. Tigrinya prepositions show behavior reminiscent of cliticization, as noted by (Kifle 2011:p. 164, fn. 3). A number of prepositions undergo contraction with a following definite determiner, and light CV prepositions attach to a following noun (24a–b).

(24) *Tigrinya prepositions undergo contraction and cliticization:*

a. እቶም ቆልዑ ምስቲ ከልቢ ይጻወቱ ኣለዉ።
ʔɨt-om k’olɨʕu **mɨs-ti** kälbi jɨ-ts’awɨt-u ʔall-ɨwu.
DEF-MP child.PL with-DEF.MS dog 3MP-play.IMPF-3MP AUX-3MP
‘The children are playing with the dog.’

[10] Kifle (2011) also cites examples with *silä* ‘for’ as a benefactive preposition.

b. **ንዮናስ ቡን ኣምጺኣሉ።**
nɨ-Jonas bun ʔa-mts'iʔ-ä-lu
DAT-Jonas coffee CAUS-bring.GER-1MS-3MS.IO
'I brought coffee for Jonas.'

Prepositions then appear to be proclitics, although there is no evidence for displacement within prepositional phrases. But we can then understand the leftward displacement of Tigrinya complementizers as the result of reanalyzing a proclitic preposition that leans on its complement as a clause-final complementizer without an obvious following host. This pattern of complementizer placement has now been generalized to non-prepositional complementizers, such as the interrogative *do* and *ʔɨntä*.[11]

As suggested above, I propose that complementizers are underlyingly head-final, like verbs or auxiliaries. There are a few other pieces of evidence for the idea that these complementizers are generated in a clause-peripheral position. First, the question particle *do* also surfaces in clause-final position (25a–b).

(25) *Question particle do can be clause-final:*

a. **ዮሰፍ ቡን ይሰቲ ዶ፧**
Josäf bun jɨ-säti **do**?
Josef coffee 3MS-drink.IMPF Q
'Does Josef drink coffee?'

b. **እቲ ቆልዓ ደቂሱ ኔሩ ዶ፧**
ʔɨt-i qolʕa däk'is-u ner-u **do**?
DEF-MS baby sleep.GER-3MS AUX.PST-3MS Q
'Had the baby slept?'

Second, the preferred placement of complementizers shows that they are oriented to the clause-final position. Proclitic complementizers always appear before a clause-final verb in clauses without auxiliaries (26a). In addition, in all verb+auxiliary combinations, the complementizer can be placed before the final auxiliary (26b–c).

(26) *Complementizers on final verb or auxiliary:*

a. **ዮሰፍ ጸጋ ትማሊ ከምዝደረፈት ሓቢሩ።**
Josäf [CP Ts'äga tɨmali **käm**=zɨ-däräf-ät] ħabir-u.
Josef Ts'äga yesterday C=REL-sing.PERF-3FS say.GER-3MS
'Josef said that Ts'äga sang yesterday.'

b. **እቲ ቆልዓ ደቂሱ ዶ ኔሩ፧**
ʔɨt-i qolʕa däk'is-u **do** ner-u?
DEF-MS baby sleep.GER-3MS Q AUX.PST-3MS
'Had the baby slept?'

c. **እቲ ህጻን ደቂሱ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።**
[CP ʔɨt-i häts'an däk'is-u **sɨlä** z-äll-o] ʔɨt-i säbʔaj ħagus ʔɨjj-u.
DEF-MS baby sleep.GER-3MS because REL-AUX-3MS DEF-MS man happy COP-3MS
'Because the baby has slept, the man is happy.'

In fact, in some verb+auxiliary combinations, complementizers only surface before the auxiliary. In the future, only the auxiliary is an eligible host, and not the verb, as illustrated for the declarative complementizer *käm* in (27a–b).

[11] The absence of any kind of displacement with true prepositions provides an argument against the idea that any of these complementizers are still synchronically prepositions. As a result, I will treat all of these items as head-final C heads.

(27) *In future, only auxiliary hosts a complementizer:*

a. ካብ ገዛ ክትወጽእ ከምዝኾነት ሓቢረ።
[CP kab gäza kɨ-t-wäts'ɨʔ **käm**=zɨ-xon-ät] ħabir-ä
to house PURP-3FS-leave.IMPF C=REL-COP.PFV-3FS report.1S
'I reported that she will leave the house.'

b. * ካብ ገዛ ከምክትወጽእ ዝኾነት ሓቢረ።
*[CP kab gäza **käm**=kɨ-t-wäts'ɨʔ zɨ-xon-ät] ħabir-ä
from house C=PURP-3FS-leave.IMPF REL-COP.PFV-3FS report.1S
'I reported that she will leave the house.'

To capture these observations, I propose that complementizers in Tigrinya are generated in head-final position, but that movement operations occur that cause them to be realized before the final verb or auxiliary. The key question now is which element undergoes movement. The effect of leftward displacement can be achieved in a variety of ways, such as upward head movement of the highest verb/auxiliary to a head-final C or by dislocation of the complementizers. In the next section, I argue that complementizers undergo an operation of lowering, on the basis of the observation that complementizers can occur before a verb in a verb+auxiliary combination.

3.2 Leftward movement of complementizers across auxiliaries

Complementizer placement in verb+auxiliary combinations provides evidence that complementizers undergo lowering, because they can move farther leftward. In clauses containing a verb and auxiliary, complementizers may sometimes attach to the verb across an auxiliary.

All three speakers consulted readily accept cases in which complementizers attach to a verb across an auxiliary, though there is variation with respect to specific examples. In the present progressive, in which the present tense auxiliary *ʔall-* combines with the imperfective stem, complementizers can appear on the verb or auxiliary (28a–d).

(28) *Present progressive permits complementizer on verb or auxiliary:*

a. ዮሰፍ ጸጋ ትደርፍ ከምዘየላ ሓቢሩ።
Josäf [CP Ts'äga tɨ-därrɨf **käm**=z-äj=äll-a] ħabir-u.
Josäf Ts'äga 3FS-sing.IMPF C=REL-NEG=AUX-3FS report-3MS
'…that Ts'äga is not singing.'

b. ዮሰፍ ጸጋ ከምዘይትደርፍ ዘላ ሓቢሩ።
Josäf [CP Ts'äga **käm**=z-äj=tɨ-därrɨf z-äll-a] ħabir-u.
Josäf Ts'äga C=REL-NEG=3FS-sing.IMPF REL-AUX-3FS report-3MS
'Josäf reported that Ts'äga is not singing.'

c. እቲ ህጻን ይድቅስ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።
[CP ʔɨt-i häts'an jɨ-dɨk'ɨs **sɨlä** z-äll-o] ʔɨt-i sɨbʔaj ħagus ʔɨjj-u.
DEF-MS baby 3MS-sleep.IMPF because REL-AUX-3MS DEF-MS man happy COP-3MS
'Because the baby is sleeping, the man is happy.'

d. እቲ ህጻን ስለ ዝድቅስ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።
[CP ʔɨt-i häts'an **sɨlä** zɨ-dɨk'ɨs z-äll-o] ʔɨt-i sɨbʔaj ħagus ʔɨjj-u.
DEF-MS baby because REL-sleep.IMPF REL-AUX-3MS DEF-MS man happy COP-3MS
'Because the baby is sleeping, the man is happy.'

Similarly, in the past progressive, in which the imperfective surfaces with the past tense auxiliary *ner-*, complementizers can appear on the final auxiliary or on the verb (29a–b).

(29) *Past progressive permits complementizers on verb or auxiliary:*

a. ካብ ገዛ ትወጽእ ከምዝነበረት ሓቢረ።
[CP kab gäza tɨ-wäts'ɨʔ **käm**=zɨ-näbär-ät] ħabir-ä
from house 3FS-leave.IMPF C=REL-AUX.PFV.PST-3FS report-1S
'I reported that she was leaving the house.'

b. ካብ ገዛ ከምትወጽእ ዝነበረት ሓቢረ።
[CP kab gäza **käm**=tɨ-wäts'ɨʔ zɨ-näbär-ät] ħabir-ä
from house C=3FS-leave.IMPF REL-AUX.PFV.PST-3FS report-1S
'I reported that she was leaving the house.'

In the present and past perfect, complementizers can also be placed before the auxiliary or before the perfective verb. The examples in (30a–b) and (31a–b) show examples of this optionality with the interrogative complementizer *ʔɨntä* 'if/whether'.

(30) *Present perfect permits complementizers on verb or auxiliary:*

a. እቲ ሰብኣይ ነታ ሰበይቲ እቲ ቆልዓ ደቂሱ እንተ ኣሎ
ʔɨt-i säbʔaj n-ät-a säbäjti [CP ʔɨt-i qolʕa däk'is-u **ʔɨntä** ʔall-o]
DEF-MS man DAT-DEF-FS woman DEF-MS child sleep.GER-3MS whether AUX-3MS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
'The man asked the woman whether the child has slept.'

b. እቲ ሰብኣይ ነታ ሰበይቲ እቲ ቆልዓ እንተ ደቂሱ ኣሎ
ʔɨt-i säbʔaj n-ät-a säbäjti [CP ʔɨt-i qolʕa **ʔɨntä** däk'is-u ʔall-o]
DEF-MS man DAT-DEF-FS woman DEF-MS child sleep.GER-3MS whether AUX-3MS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
'The man asked the woman whether the child has slept.'

(31) *Past perfect permits complementizers on verb or auxiliary:*

a. እቲ ሰብኣይ ነታ ሰበይቲ እቲ ቆልዓ ደቂሱ እንተ ኔሩ
ʔɨt-i säbʔaj n-ät-a säbäjti [CP ʔɨt-i qolʕa däk'is-u **ʔɨntä** ner-u]
DEF-MS man DAT-DEF-FS woman DEF-MS child sleep.GER-3MS whether AUX.PST-3MS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
'The man asked the woman whether the child had slept.'

b. እቲ ሰብኣይ ነታ ሰበይቲ እቲ ቆልዓ እንተ ደቂሱ ኔሩ
ʔɨt-i säbʔaj n-ät-a säbäjti [CP ʔɨt-i qolʕa **ʔɨntä** däk'is-u ner-u]
DEF-MS man DAT-DEF-FS woman DEF-MS child sleep.GER-3MS whether AUX.PST-3MS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
'The man asked the woman whether the child had slept.'

Examples in which the complementizers appears before the clause-final auxiliary are always accepted. There is considerable variation between and within speakers as to whether attachment to the verb is possible in any given pair, and there seems to be an ameliorating effect in some examples if the negative clitic *ʔaj*= attaches to the verb also.[12] But all speakers accept cliticization to the verb in a range

[12] The negative clitic is typically hosted by the verb (though it can appear on the auxiliary in some verb+auxiliary combinations), and there may be a preference to form a 'clitic cluster'.

of cases, with and without negation. I conclude that non-local cliticization to the verb across an auxiliary is in principle possible in all tense-aspect combinations described above.

One exception to this pattern, as noted in the previous section, may arise in the construction that expresses the future. The future is formed by means of the copula *ʔɨjj-* and the imperfective stem prefixed with the purposive prefix *kɨ-*. Complementizers must be on the auxiliary (32a–b).

(32) *In future, only auxiliary hosts a complementizer:*

a. ካብ ገዛ ክትወጽእ ከምዝኾነት ሓቢረ።
[$_{\text{CP}}$ kab gäza kɨ-t-wäts'ɨʔ **käm**=zɨ-xon-ät] ħabir-ä
to house PURP-3FS-leave.IMPF C=REL-COP.PFV-3FS report.1S
'I reported that she will leave the house.'

b. * ካብ ገዛ ከምክትወጽእ ዝኾነት ሓቢረ።
*[$_{\text{CP}}$ kab gäza **käm**=kɨ-t-wäts'ɨʔ zɨ-xon-ät] ħabir-ä
from house C=PURP-3FS-leave.IMPF REL-COP.PFV-3FS report.1S
'I reported that she will leave the house.'

I suggest that the future makes use of a biclausal structure, in which the copula embeds an infinitival clause. This clause boundary prevents displacement of the complementizer onto the verb.

Non-local cliticization of complementizers provides an important argument that complementizer placement reflects an operation of lowering. I will develop an analysis in which lowering can occur successively, so that a head-final complementizer can cliticize first onto the final auxiliary (33) and then optionally onto the lower verb (34).

(33) *Lowering onto the auxiliary:*
[$_{\text{V}}$ tɨ-wäts'ɨʔ$_{\text{V}}$] [$_{\text{T}}$ **käm**= [$_{\text{T}}$ zɨ-näbär-ät$_{\text{Aux}}$]] käm=

(34) *Successive lowering onto the verb:*
[$_{\text{V}}$ **käm**= [$_{\text{V}}$ tɨ-wäts'ɨʔ$_{\text{V}}$] [$_{\text{T}}$ käm= [$_{\text{T}}$ zɨ-näbär-ät$_{\text{Aux}}$]] käm=

Non-local cliticization is more difficult to handle in an analysis that makes use of upward head movement to C. The verb and auxiliary are independent words, each with their own prefixes, suffixes, and vocalic melody, and so only the highest verb/auxiliary should raise to C. In addition, we will see in section 5 that Tigrinya provides evidence for the intermediate step of lowering in (34), since the auxiliary still shows the morphological effects associated with the complementizer.

A second argument for lowering over upward head movement comes from constructions in which the highest verb or auxiliary is not final. Tigrinya has a cleft construction in which the copula *ʔɨjj-* may appear in between the pivot and a relative clause (35a). When such a cleft is embedded, the complementizer may surface in clause-medial position, lowering onto the copula (35b).

(35) *Lowering to clause-medial copula:*

a. ዮሰፍ እዩ ካብ ገዛ ዝወጸአ።
Josäf ʔɨjj-u kab gäza zɨ-wäts'äʔ-ä
Josäf COP-3MS from house REL-leave.PERF-3MS
'It is Yosef who left the house.'

b. ዮሰፍ ከምዝኾነ ካብ ገዛ ዝወጸ ሰሚዐ።
[$_{\text{CP}}$ Josäf **käm**=zɨ-xon-ä kab gäza zɨ-wäts'äʔ-ä] sämiʕ-ä.
Josäf C=REL-COP.PFV-3MS from house REL-leave.PERF-3MS hear-1S
'I heard that it is Yosef who left the house.'

This possibility follows from a lowering analysis, but would require extraposition of the relative clause to a high adjoined position in an analysis relying on head movement.

I conclude that Tigrinya complementizer placement reflects leftward displacement of the complementizers into the head-final cluster. In the next section, I develop an analysis of this pattern using Embick and Noyer's (2001) postsyntactic operation of Lowering.

4 A Lowering analysis of Tigrinya complementizers

In this section, I argue for an account of Tigrinya complementizer placement as the outcome of a postsyntactic operation of Lowering (Embick and Noyer 2001; Arregi and Pietraszko 2021; Georgieva et al. 2021). To support a morphosyntactic conception of lowering, I demonstrate that complementizer placement affects allomorphy and triggers a morphological rule of *zɨ*-insertion, which I argue reflects a rule of disassociated node insertion in the sense of Noyer (1998).

4.1 A postsyntactic rule of Lowering

I adopt a model of grammar in which the output of syntax is interpreted by a postsyntactic morphological component, in which morphological rules and rules of exponence apply, prior to evaluation by a phonological grammar (Halle and Marantz 1993 et seq.). Within this morphological module, morphological rules like Impoverishment, Fission, or Lowering may apply, driven by language-specific morphological wellformedness requirements. I argue that complementizer placement in Tigrinya reflects the application of a rule of Lowering in the sense of Embick and Noyer (2001) (see also Harizanov and Gribanova 2019; Georgieva et al. 2021; Weisser 2024). Embick and Noyer conceive of Lowering as a morphological operation that takes two syntactic heads X and Y in the output generated by syntax and adjoins the higher head X to the lower head Y, as schematized in (36).

(36) **Lowering (Embick and Noyer 2001:561):**
$[_{\text{XP}}\ X^0 \ldots [_{\text{YP}} \ldots Y^0 \ldots]] \rightarrow [_{\text{XP}} \ldots [_{\text{YP}} \ldots [_{Y^0}\ Y^0{+}X^0] \ldots]]$

This rule converts a relation of hierarchical adjacency (the head-complement relationship) into one of linear adjacency. Importantly, this rule applies to a hierarchical representation, prior to the application of Vocabulary Insertion, which associates terminals with their phonological content.

I propose that Lowering in Tigrinya is driven by a morphological constraint that applies to a subset of complementizers to the effect that they must immediately precede a head bearing a category feature [V] within a complex head. I state this wellformedness requirement in (37) (see also Embick and Noyer 2001:586 for a similar filter for English Lowering).

(37) *Morphological requirement on Tigrinya complementizers:*
C must be in the context $[_{X^0}$ ___ $X^0_{[\text{CAT:V}]}]$

Abstract syntax is blind to morphological filters of this type and generates a strict head-final clause. Complementizer displacement is not driven by a syntactic feature that is visible to syntax and so no syntactic movement applies.[13] But, in the postsyntactic component, the requirement in (37) will require resolution, and so trigger the application of Lowering.

I demonstrate this logic with a derivation for the example in (38a). The underlying syntax for the embedded CP is head-final and is represented in (38b). I annotate the tree with overt morphemes for ease of exposition, but note that the syntax manipulates abstract feature bundles only.

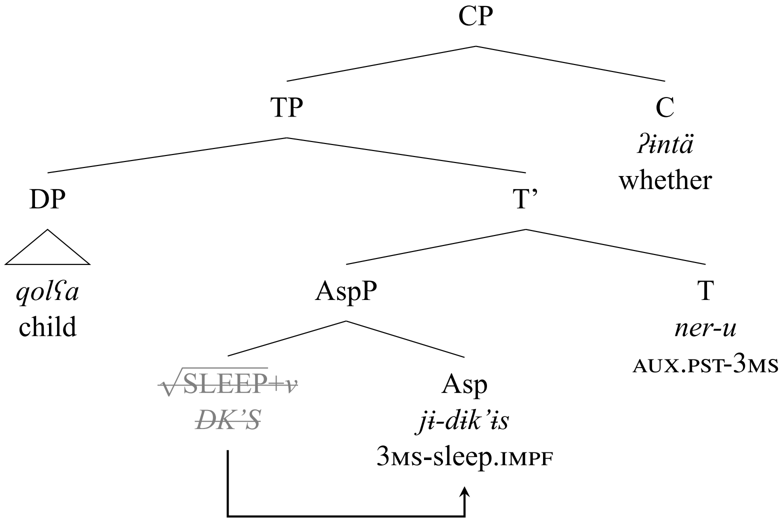
(38) *Abstract syntax for embedded CP in (38a):*

a.	**እቲ**	**ሰብኣይ**	**ነታ**	**ሰባይቲ**	**ቆልዓ**	**ይድቅስ**	**እንተ**	**ኔሩ**
	ʔɨt-i	säbʔaj	n-ät-a	säbajti	[$_{\text{CP}}$ qolʕa	jɨ-dɨk'ɨs	**ʔɨntä**	ner-u]
	DEF-MS	man	DAT-DEF-FS	woman	child	3MS-sleep.IMPF	whether	AUX.PST-3MS

[13] Another possibility is that there are features that only generate displacement in the morphology, like the [+M] or [-M] features proposed by Harizanov and Gribanova (2019) or the integration features developed by Weisser (2024). I stick to an account here that avoids purely morphological features.

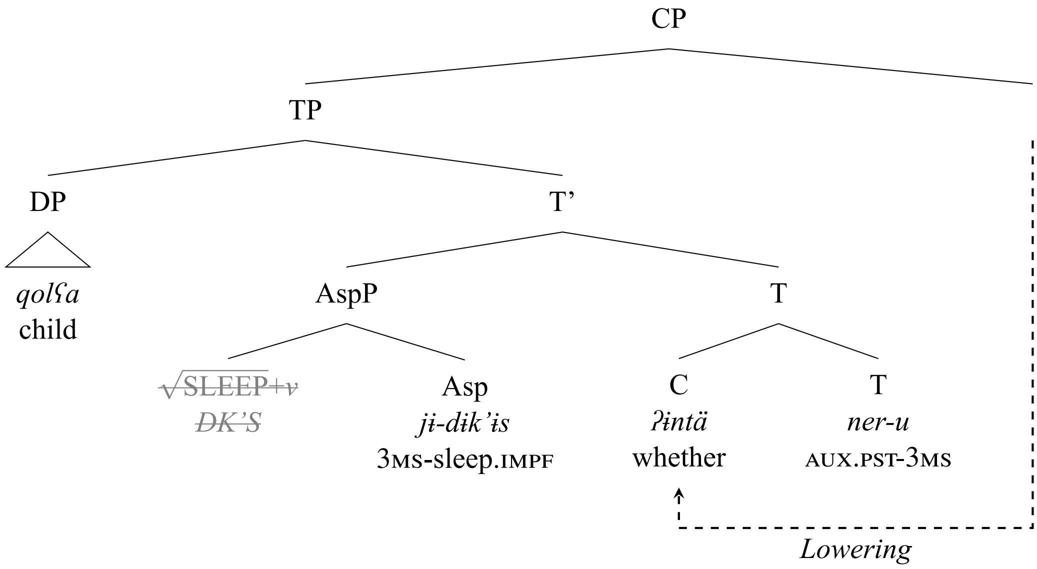
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
‘The man asked the woman whether a child was sleeping.’

b.



The verb undergoes head movement to Asp in the syntax, but the complementizer is not targeted by syntactic movement operations. When this syntax is interpreted in the morphological component, the wellformedness requirement on complementizers in (37) triggers a rule of Lowering. Lowering applies prior to Vocabulary Insertion and adjoins the feature bundle that instantiates the interrogative complementizer to T (39).

(39) *Postsyntactic Lowering in embedded CP in (38a):*



If the verb moves to T instead, in clauses without an auxiliary, the verb will host the lowered complementizer.

Embick and Noyer’s (2001) operation of Lowering then captures the preferred pattern of complementizer placement in Tigrinya, which targets the last verb or auxiliary in the clause. Since the complementizer is displaced from a clause-final position, it is generally lowered onto the closest head-final verbal element.

4.2 Complementizer placement and relative clause morphology

One key question about lowering operations is whether these ought to be handled with morphosyntactic rules (Halle and Marantz 1993), or with phonological rules that invert linearly adjacent words, similar to rules of infixation or metathesis, like Halpern's (1995) Prosodic Inversion or Embick and Noyer's (2001) Local Dislocation.

Complementizer placement in Tigrinya has two related morphological effects that demonstrate it is morphosyntactic in nature. First, embedded complementizers that are prepositional in origin trigger insertion of the relative prefix *zɨ-* (underlined) on eligible hosts. The examples in (40a–b) demonstrate for the declarative complementizer *käm* and the adverbial complementizer *sɨlä*.

(40) *Complementizers trigger insertion of relative prefix zɨ-: on finite verb*

a. **ዮሰፍ ጸጋ ትማሊ ከምዝደረፈት ሓቢሩ።**
Josäf [$_{\text{CP}}$ Ts'äga tɨmali **käm**=zɨ-däräf-ät] ħabir-u.
Josef Ts'aiga yesterday c=rel-sing.perf-3fs say.ger-3ms
'Josef said that Ts'aiga sang yesterday.'

b. **እቲ ህጻን ስለ ዝደቀሰ እቲ ሰብኣይ ትሓጐሱ።**
[$_{\text{CP}}$ ʔɨt-i häts'an **sɨlä** zɨ-däk'äs-ä] ʔɨt-i säbʔaj tɨ-ħagwis-u.
def-ms baby because rel-sleep.perf-3ms def-ms man nact-happy.ger-3ms
'Because the baby has slept, the man is happy.'

The same effect obtains when a clause-final auxiliary is the host (41a–b).

(41) *Complementizers trigger insertion of relative prefix zɨ- on finite auxiliary:*

a. **ካብ ገዛ ወጺኣ ከምዘላ ሓቢረ።**
[$_{\text{CP}}$ kab gäza wäts'iʔ-a **käm**=z-äll-a] ħabir-ä
from house leave.ger-3fs c=rel-aux-3fs report.ger-1s
'I reported that she has left the house.'

b. **እቲ ህጻን ደቂሱ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።**
[$_{\text{CP}}$ ʔɨt-i häts'an däk'is-u **sɨlä** z-äll-o] ʔɨt-i säbʔaj ħɨgus ʔɨjj-u.
def-ms baby sleep.ger-3ms because rel-aux-3ms def-ms man happy cop-3ms
'Because the baby has slept, the man is happy.'

The *zɨ-* prefix otherwise marks relative clauses (Leslau 1941:61–63; Palmer 1962). Its distribution is linked to the prefixes carried by the verb it attaches to. The relative prefix appears on verbs without other prefixes. Perfective verbs, which lack agreement prefixes, host relative morphology (42a). On imperfective verbs, the relative prefix seems to coalesce with the agreement prefix *jɨ-* (42b), but is absent before other agreement prefixes like *tɨ-* (42c).[14]

(42) *The zɨ- prefix marks relative clauses on verbs:*

a. **ነሕና ዝሰተናዮ ቡን መሪር እዩ።**
[$_{\text{CP}}$ näħɨna **zɨ**-sätä-na-jo] bun märrir ʔɨjj-u
1pl rel-drink.perf-1pl-3ms.o coffee bitter cop-3ms
'The coffee that we drank tastes bitter.'

b. **እቲ ቡን ዝሰቲ ወዲ ሕጉስ እዩ።**
ʔɨt-i [$_{\text{CP}}$ bun **zɨ**-säti] wäddi ħɨgus ʔɨjj-u.
def-ms coffee rel.3ms-drink.impf boy happy cop-3fs
'The boy that drinks coffee is happy.'

[14]Leslau (1941:61–63) reports relative morphology before imperfective agreement prefixes, but my consultants drop *zɨ-* before the agreement prefixes *tɨ-* and *nɨ-* and Cacchioli (2023) reports the same pattern for the speakers she works with.

c. እታ ቡን ትሰቲ ጓል ሕጉስቲ እያ።
ʔɨt-a [$_{CP}$ bun **tɨ**-säti] gwal ħɨgus-ti ʔɨjj-a.
DEF-FS coffee 3FS-drink.IMPF girl happy-FEM COP-3FS
'The girl that drinks coffee is happy.'

In verb+auxiliary combinations, the auxiliary must be marked with the relative prefix. The verb optionally carries this marking (43a–b).[15]

(43) *Relative prefix optional on verbs in verb+auxiliary combinations:*

a. እታ ትማሊ ቡን ዝሰተየት ዝነበረት ጓል ሕጉስ
ʔɨt-a [$_{CP}$ tɨmali bun **zɨ**-sätäj-ät **zɨ**-näbär-ät] gwal ħɨgus
DEF-FS yesterday coffee REL-drink.PERF-3FS REL-AUX.PFV.PST-3FS girl happy
ነራ።
ner-a.
AUX.PST-3FS
'The girl that had drunk coffee yesterday was happy.'

b. እታ ትማሊ ቡን ሰቲያ ዝነበረት ጓል ሕጉስ ነራ።
ʔɨt-a [$_{CP}$ tɨmali bun sätij-a **zɨ**-näbär-ät] gwal ħɨgus nejr-a.
DEF-FS yesterday coffee drink.GER-3FS REL-AUX.PFV.PST-3FS girl happy AUX.PST-3FS
'The girl that had drunk coffee yesterday was happy.'

I propose to treat the appearance of the relative prefix *zɨ-* after complementizers as an instance of "ornamental" morphology, triggered by a morphological rule associated with certain complementizers. Although the use of relative morphology is clearly linked to the prepositional origin of some of Tigrinya's complementizers, there are a couple of pieces of evidence that relative morphology is not syntactically or semantically meaningful. First, not all complementizers are accompanied by relative clause morphology, and so there is no requirement for embedded clauses to be formed this way. The question particle *do* and the interrogative complementizer *ʔɨntä* do not trigger insertion of a relative prefix (44a–b).

(44) *No relative prefix after interrogative complementizers:*

a. *እቲ ቆልዓ ደቂሱ ዶ ዝነበረ፧
*ʔɨt-i qolʕa däk'is-u **do** zɨ-näbär-ä?
DEF-MS baby sleep.GER-3MS Q AUX.PST-3MS
'Had the baby slept?'

b. *እቲ ሰብኣይ ነታ ሰባይቲ ቆልዓ ይድቅስ እንተ ዝነበረ
*ʔɨt-i säbʔaj n-ät-a säbajti [$_{CP}$ qolʕa jɨ-dɨk'ɨs **ʔɨntä** zɨ-näbär-ä]
DEF-MS man DAT-DEF-FS woman child 3MS-sleep.IMPF whether REL-AUX.PFV.PST-3MS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
'The man asked the woman whether the baby was sleeping.'

Second, in clauses with complementizers that trigger insertion, the distribution of the relative prefix is strictly determined by the placement of the complementizer. The complementizer host is obligatorily marked by the relative prefix (45a–b).

[15]The fact that verbs can lack relative morphology when accompanied by an auxiliary is first noted for perfective verbs by Palmer (1962) (see also Cacchioli 2023). In the imperfective, relative morphology is often suppressed on verbs, as in (42c), so it is difficult to tell. Imperfective verbs with 1SG/3MS subjects seem to show relative morphology obligatorily in relative clauses (Palmer 1962; see Cacchioli 2023).

(45) *Complementizers must be accompanied by relative prefix zɨ-:*

a. *ዮሰፍ ጻጋ ትማሊ ከምደሪፉ ሓቢሩ።
*Josäf [$_{CP}$ Ts'äga tɨmali **käm**=därif-a] ħabir-u.
Josef Ts'äga yesterday C=sing.GER-3FS say.GER-3MS
'Josef said that Ts'äga sang yesterday.'

b. * እቲ ህጻን ስለ ደቂሱ እቲ ሰብኣይ ትሓጉሱ።
*[$_{CP}$ ʔɨt-i häts'an **sɨlä** däk'is-u] ʔɨt-i säbʔaj tɨ-ħagwis-u.
DEF-MS baby because sleep.GER-3MS DEF-MS man NACT-happy.GER-3MS
'Because the baby has slept, the man is happy.'

In addition, verbs not preceded by the complementizer cannot be marked by the prefix (46a–b).

(46) *Relative prefix impossible on verbs that do not host a complementizer:*

a. ካብ ገዛ ዝወጸአ ከምዘሎ ሓቢረ።
*[$_{CP}$ kab gäza z̲ɨ̲-wäts'äʔ-ä **käm**=z̲-äll-o] ħabir-ä
from house REL-leave.PERF-3MS C=REL-AUX-3MS report-1S
'I reported that he has left the house.'

b. * እቲ ህጻን ዝደቀሰ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ
*[$_{CP}$ ʔɨt-i häts'an z̲ɨ̲-däk'äs-ä **sɨlä** z̲-äll-o] ʔɨt-i säbʔaj ħɨgus
DEF-MS baby REL-sleep.PERF-3MS because REL-AUX-3MS DEF-MS man happy
እዩ።
ʔɨjj-u.
COP-3MS
'Because the baby has slept, the man is happy.'

But, in cases of non-local cliticization, when the complementizer precedes the verb, both verb and auxiliary are marked by the relative prefix (47a–b).

(47) *Relative prefix on verb and auxiliary in non-local cliticization:*

a. ዮሰፍ ጻጋ ከምዘይትደርፍ ዘላ ሓቢሩ።
Josäf [$_{CP}$ Ts'äga **käm=**z̲-äj=tɨ-därrɨf z̲-äll-a] ħabir-u.
Josäf Ts'äga C=REL-NEG=3FS-sing.IMPF REL-AUX-3FS report-3MS
'Josäf reported that Ts'äga is not singing.'

b. እቲ ህጻን ስለ ዝድቅስ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።
[$_{CP}$ ʔɨt-i häts'an **sɨlä** z̲ɨ̲-dɨk'ɨs z̲-äll-o] ʔɨt-i sɨbʔaj ħɨgus ʔɨjj-u.
DEF-MS baby because REL-sleep.IMPF REL-AUX-3MS DEF-MS man happy COP-3MS
'Because the baby is sleeping, the man is happy.'

In other words, the spreading of relative morphology follows the displacement of complementizers. It seems to be a requirement of the relevant complementizers then that they "mark" their host with vacuous relative morphology. To capture this observation, I propose that such complementizers trigger a rule of *disassociated node insertion*, in the sense of Noyer (1998). I call this rule Rel Insertion, and it inserts a relative node on any verbal head that is preceded by a declarative complementizer (48).

(48) **Rel Insertion:**
Insert a node Rel in the context $[C_{DECL} [\_\_\_ X_{[CAT:V]}]]$

A morphological rule of Rel Insertion explains why the relative prefix is associated with some complementizers, but does not seem to be necessary to construct an embedded clause. In addition, treating the distribution of *zɨ-* as a purely morphological fact explains why this prefix cannot appear on verbs that do not host a complementizer.

The rule of Rel Insertion must follow the application of Lowering, to ensure that the host of the complementizer is obligatorily marked by relative morphology. In fact, when we turn to examples with multiple steps of Lowering, we will see that the application of Rel Insertion is interleaved in between different applications of Lowering. This morphological effect can be understood if Lowering is morphosyntactic, and not purely phonological displacement.

4.3 Stem allomorphy and complementizers

The second morphological effect associated with complementizer displacement is the allomorphy pattern found in perfective stems, as described in section 2.1. Displaced complementizers and the relative prefix associated with them trigger the perfective allomorph of their host, allowing us to demonstrate that the application of Lowering must feed exponent choice, as in a morphosyntactic conception of Lowering.

Recall from section 2.1 that the perfective stem in Tigrinya is typically indicated by the gerund allomorph, which is associated with a unique vocalic melody and set of agreement suffixes. Negation and declarative complementizers trigger an allomorphy effect, so that the perfective allomorph appears instead. As a result, the complementizers *käm* and *sɨlä* 'because' require the perfective allomorph when they are hosted by a perfective verb (49a–b).

(49) *Complementizers trigger perfective allomorph of verb:*

a. **ዮሰፍ ጸጋ ትማሊ ከምዝደረፈት ሓቢሩ።**
Josäf [$_{CP}$ Ts'äga tɨmali **käm**=zɨ-däräf-ät] ħabir-u.
Josef Ts'aiga yesterday C=REL-sing.PERF-3FS say.GER-3MS
'Josef said that Ts'aiga sang yesterday.'

b. **እቲ ህጻን ስለ ዝደቀሰ እቲ ሰብኣይ ትሓጐሱ።**
[$_{CP}$ ʔɨt-i häts'an **sɨlä** zɨ-däk'äs-ä] ʔɨt-i säbʔaj tɨ-ħagwis-u.
DEF-MS baby because REL-sleep.PERF-3MS DEF-MS man NACT-happy.GER-3MS
'Because the baby has slept, the man is happy.'

The same allomorphy effect is found with auxiliaries, with the past tense auxiliary *ner*- and the stative copula *ʔɨjj*-. These auxiliaries surface with a perfective allomorph when they host an appropriate complementizer (50a–b). These allomorphs display some irregularity, but can be recognized as perfective, because they trigger the perfective agreement suffixes.

(50) *Complementizer triggers perfective allomorph of auxiliary:*

a. **ካብ ገዛ ትወጽእ ከምዝነበረት ሓቢረ።**
[$_{CP}$ kab gäza tɨ-wäts'ɨʔ **käm**=zɨ-näbär-ät] ħabir-ä
from house 3FS-leave.IMPF C=REL-AUX.PFV.PST-3FS report-1S
'I reported that she was leaving the house.'

b. **ካብ ገዛ ክትወጽእ ከምዝኾነት ሓቢረ።**
[$_{CP}$ kab gäza kɨ-t-wäts'ɨʔ **käm**=zɨ-xon-ät] ħabir-ä
to house PURP-3FS-leave.IMPF C=REL-COP.PFV-3FS report.1S
'I reported that she will leave the house.'

This effect is an instance of allomorphy, because it requires the complementizer to linearly precede the verb. Perfective stems that do not host these complementizers are realized in the gerund form (51a–b).

(51) *Gerund allomorph appears when not hosting complementizer:*

a. **ካብ ገዛ ወጺኣ ከምዘላ ሓቢረ።**
[$_{CP}$ kab gäza wäts'iʔ-a **käm**=z-äll-a] ħabir-ä
from house leave.GER-3FS C=REL-AUX-3FS report.GER-1S
'I reported that she has left the house.'

b. **እቲ ህጻን ደቂሱ ስለ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።**
[$_{CP}$ ʔɨt-i häts'an däk'is-u **sɨlä** z-äll-o] ʔɨt-i säbʔaj ħɨgus ʔɨjj-u.
DEF-MS baby sleep.GER-3MS because REL-AUX-3MS DEF-MS man happy COP-3MS
'Because the baby has slept, the man is happy.'

As evident in the forms above, perfective stem allomorphy co-occurs with insertion of the relative prefix. Indeed, perfective allomorphs appear in relative clauses as well. I propose that it is the relative prefix that acts as the allomorphy trigger in all cases. It is worth noting, though, that there are instances in which the relative prefix appears to be suppressed, but stem allomorphy still occurs. For example, the complementizer *mɨs* 'when' does not trigger insertion of relative clause morphology, but does condition the perfective allomorph (52a–b).

(52) *The complementizer mɨs triggers perfective allomorph but not relative prefix:*

a. እቲ ህጻን ምስ ደቀሰ እቲ ሰብኣይ ተሓጎሱ።
[CP ʔɨt-i häts'an **mɨs** <u>däk'äs-ä</u>] ʔɨt-i säbʔaj tɨ-ħagwis-u
DEF-MS baby when sleep.PERF-3MS DEF-MS man NACT-happy.GER-3MS
'When the baby slept, the man became happy.'

b. * እቲ ህጻን ምስ ዝደቀሰ እቲ ሰብኣይ ተሓጎሱ።
*[CP ʔɨt-i häts'an **mɨs** zɨ-<u>däk'äs-ä</u>] ʔɨt-i säbʔaj tɨ-ħagwis-u
DEF-MS baby when REL-sleep.PERF-3MS DEF-MS man NACT-happy.GER-3MS
'When the baby slept, the man became happy.'

I treat such cases as allomorphy of the Rel node, and will continue to take the relative prefix to be the allomorphy trigger.[16]

Stem allomorphy provides a clear argument for the idea that Lowering in Tigrinya is morphosyntactic, and does not reflect phonological reordering. A Lowering analysis captures the feeding relationship between complementizer placement and allomorphy, on the assumption that rules of Lowering precede Vocabulary Insertion, as in the Distributed Morphology architecture proposed by Embick and Noyer (2001). Consider the derivation of an example with the complementizer *käm*, with associated *zɨ-* insertion and stem allomorphy (53a). In the postsyntax, a rule of Lowering applies to adjoin the complementizer to T (53b).

(53) *Lowering in the postsyntactic derivation of (53a):*

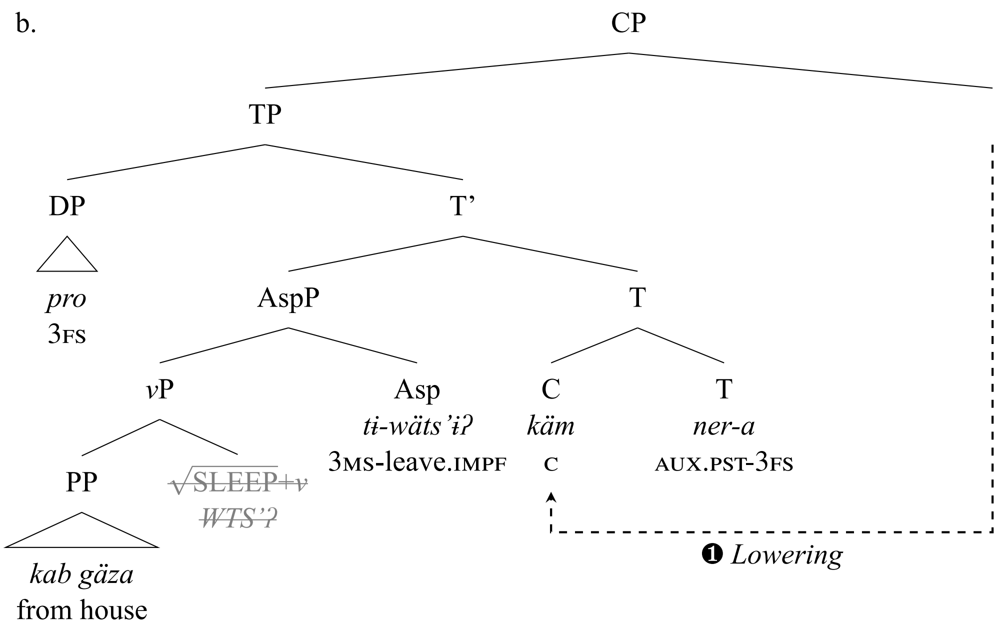
a. ካብ ገዛ ትወጽእ ከምዝነበረት ሓቢረ።
[CP kab gäza tɨ-wäts'ɨʔ **käm**=zɨ-<u>näbär-ät</u>] ħabir-ä
from house 3FS-leave.IMPF C=REL-AUX.PFV.PST-3FS report-1s
'I reported that she was leaving the house.'

[16]Evidence for this conclusion comes from the fact that *mɨs* 'when' does seem to tolerate relative morphology on an imperfective base that displays relative morphology, such as one with a 3MS agreement prefix (see (42a–c)). The example in (i) demonstrates.

(i) *The complementizer mɨs allows relative morphology on imperfective:*

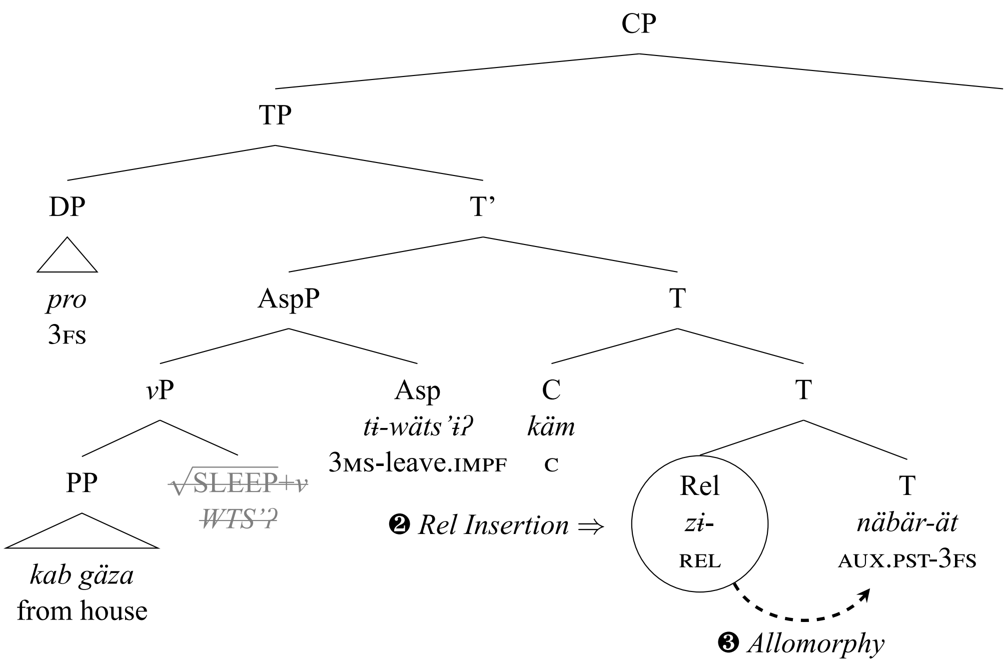
እቲ ቖልዓ ምስ ዝድቅስ ዮሰፍ መጽሓፍ የንብብ።
ʔɨt-i qolʕa mɨs zɨ-dɨk'ɨs Josäf mäts'ħaf j-änɨbɨb
DET-3MS baby when REL.3MS-sleep.IMPF Josäf book 3MS-read.IMPF
'When the baby sleeps, Josäf reads a book.'

b.



The rule of Rel Insertion then applies, to adjoin a Rel node to T, as schematized in (54).

(54) *Rel Insertion in the postsyntactic derivation of (53a):*



Subsequent to Lowering and Rel Insertion, Vocabulary Insertion applies. The relative node prefixed to T now conditions the perfective stem allomorph of the complementizer host, in this case the *näbär-* stem of the past tense auxiliary and the associated agreement suffixes, as represented in the structure in (54).

In the next section, I extend this analysis to cases of non-local cliticization, which I argue arise by successive Lowering. Evidence for this conclusion comes from the fact that the morphological rules described above still apply in intermediate positions, on auxiliaries crossed by cliticization, reflecting a derivation in which the application of rules of Lowering and node insertion is interleaved.

5 Successive Lowering

This section develops an account of non-local cliticization in Tigrinya and argues that it arises by means of successive steps of Lowering. First, I present evidence that non-local cliticization involves two steps of Lowering, from the observation that Tigrinya displays reflexes of Lowering in intermediate position, in the form of stem allomorphy and the insertion of relative morphology. On this basis, I propose a conception of Embick and Noyer's (2001) Lowering in which it is not limited to heads in a head-complement relation, but simply targets the closest head (see also Adamson 2022; Weisser 2024). If excorporation of clitics is possible (Roberts 1991), then this view can deliver non-local cliticization. I then demonstrate how the distribution of intermediate reflexes of Lowering follows from a serial view of the postsyntactic component (Halle and Marantz 1993; Arregi and Nevins 2012).

5.1 Intermediate reflexes of Lowering

This section argues that non-local cliticization in Tigrinya reflects the application of multiple steps of Lowering. The primary source of evidence for this conclusion comes from the distribution of the morphological effects associated with complementizers, stem allomorphy and insertion of relative morphology. As we will see, these morphological effects show up in intermediate position, offering evidence for intermediate steps of Lowering.

When a complementizer attaches to the verb across an auxiliary, the auxiliary shows evidence of an intermediate step of Lowering. In (55a–b), we see that both the verb and the auxiliary are marked with the relative prefix *zɨ*- when the verb hosts the complementizer.

(55) *Relative morphology on verb and auxiliary:*

a. ዮሰፍ ጸጋ ከምዘይትደርፍ ዘላ ሓቢሩ።
Josäf [CP Ts'äga **käm**=*z*-äj=tɨ-därrɨf *z*-äll-a] ħabir-u.
Josäf Ts'äga c=rel-neg=3fs-sing.impf rel-aux-3fs report-3ms
'Josäf reported that Ts'äga is not singing.'

b. እቲ ህጻን ስለ ዝድቅስ ዘሎ እቲ ሰብኣይ ሕጉስ እዩ።
[CP ʔɨt-i häts'an **sɨlä** *zɨ*-dɨk'ɨs *z*-äll-o] ʔɨt-i sɨbʔaj ħagus ʔɨjj-u.
def-ms baby because rel-sleep.impf rel-aux-3ms def-ms man happy cop-3ms
'Because the baby is sleeping, the man is happy.'

Similarly, complementizers that trigger stem allomorphy require the perfective allomorph of auxiliaries, even when they cliticize to the auxiliary. The complementizer *käm*, for instance, triggers the perfective allomorph of the past tense auxiliary *ner*-. This allomorph must appear, even if the complementizer attaches to the verb (56a–b).[17]

(56) *Stem allomorphy of auxiliary obligatory in non-local cliticization:*

a. ካብ ገዛ ከምዘይትወጽእ ዝነበረት ሓቢረ።
[CP kab gäza **käm**=z-äj=tɨ-wäts'ɨʔ zɨ-näbär-ät] ħabir-ä
from house c=rel-neg=3fs-leave.impf rel-aux.pfv.pst-3fs report-1s
'I reported that she was leaving the house.'

[17] Interestingly, although stem allomorphy is obligatory, the spreading of relative morphology in intermediate position appears to be optional for the past tense auxiliary *ner*-, so that (ii) is also possible.

(ii) *Relative prefix optional on intermediate näbär-:*

ካብ ገዛ ከምዘይትወጽእ ነበረት ሓቢረ።
[CP kab gäza **käm**=*z*-äj=tɨ-wäts'ɨʔ näbär-ät] ħabir-ä
from house c=rel-neg=3fs-leave.impf aux.pfv.pst-3fs report-1s
'I reported that she was leaving the house.'

I take this optionality to reflect variation in the realization of the Rel node.

b. * ካብ ገዛ ከምዘይትወጽእ ኔሩ ሓቢረ።
*[$_{\text{CP}}$ kab gäza **käm**=z-äj=tɨ-wäts’ɨʔ ner-u] ħabir-ä
from house C=REL-NEG=3FS-leave.IMPF AUX.PST-3FS report-1S
‘I reported that she was leaving the house.’

These facts also demonstrate that the verb and auxiliary behave like independent morphological words, which take prefixes independently and are associated with independent vocalic melodies. In fact, it is possible in Tigrinya for some auxiliaries to surface in a contracted form, as an enclitic onto the verb. Both the present tense auxiliary *ʔall-* and copula *ʔɨjj-* can appear as enclitics (57a–d), orthographically represented with an apostrophe.

(57) *Reduction and encliticization of ʔall- and ʔɨjj-:*

a. ካብ ገዛ ወጺኡ ኣሎ።
kab gäza wäts’iʔ-u **ʔall**-o
from house leave.GER-3MS AUX-3MS
‘He has left the house.’

b. ካብ ገዛ ወጺኡ'ሎ።
kab gäza wäts’iʔ-u=**ll**-o
from house leave.GER-3MS=AUX-3MS
‘He has left the house.’

c. እቲ ቡን መሪር እዩ።
ʔɨt-i bun märrir **ʔɨjj**-u.
DEF-MS coffee bitter COP-3MS
‘The coffee is bitter.’

d. እቲ ቡን መሪር'ዩ።
ʔɨt-i bun märrir=**jj**-u.
DEF-MS coffee bitter=COP-3MS
‘The coffee is bitter.’

This kind of contraction may be an alternative way of satisfying the phonological requirement in Tigrinya that every word has an onset consonant. These auxiliaries are arguably onsetless and otherwise surface with an epenthesized glottal stop.

But contraction does not seem to play a role in non-local cliticization. Contraction appears to remain possible in embedded clauses with complementizers that do not trigger relative morphology, but becomes ungrammatical with complementizers that do trigger spreading of relative morphology and so alter the form of the auxiliary (58b).

(58) *Contraction does not facilitate non-local cliticization:*

a. እቲ ሰብኣይ ነታ ሰበይቲ ካብ ገዛ ወጺኣ እንተ'ላ]
ʔɨt-i säbʔaj n-ät-a säbäjti [$_{\text{CP}}$ kab gäza wäts’iʔ-a **ʔɨntä=ll-a**]
DEF-MS man DAT-DEF-FS woman from house leave.GER-3FS whether=AUX-3FS
ሓቲቱዋ።
ħatit-uw-a.
ask.GER-3MS-3FS.O
‘The man asked the woman whether she has left.’

b. * ካብ ገዛ ከምዝወጸአ'ሎ ሓቢረ።
*[$_{\text{CP}}$ kab gäza **käm**=zɨ-wäts’äʔ-ä=ll-o] ħabir-ä
from house C=REL-leave.PERF-3MS=AUX-3MS report-1S
‘I reported that he has left the house.’

I conclude that cliticization is genuinely non-local and does not reflect a process of reanalysis by which the verb and auxiliary form a single prosodic word.

I propose that intermediate morphological effects arise because non-local cliticization involves an intermediate step of Lowering, in which the complementizer is hosted by the auxiliary (59).

(59) *Successive lowering onto the verb:*
[$_V$ **käm**= [$_V$ tɨ-wäts'ɨʔ$_V$] [$_T$ käm= [$_T$ zɨ-näbär-ät$_{Aux}$]] käm=

At this intermediate stage, the rule of Rel Insertion may apply and insert relative morphology. In addition, this intermediate step can explain the distribution of allomorphy, since it creates a configuration within which the complementizer is adjacent to the auxiliary. In the next section, I develop a conception of Lowering which derives the distribution of these effects.

5.2 Excorporation and successive Lowering

The Tigrinya facts described in this paper provide evidence that lowering may be non-local and does not necessarily target the closest verbal head. This view is at odds with Embick and Noyer's (2001) initial formulation of Lowering, in which Lowering translates a head-complement relation into morphological adjacency. But, based on the Tigrinya pattern, it cannot be the case that Lowering can only target the head of the complement.

In fact, Embick and Noyer (2001:589) already challenge this idea based on a discussion of Lowering of English tense and how it interacts with constituent negation, suggesting that the locality conditions on Lowering ought to be stated in terms of the closest head rather than strictly the head of the complement. Weisser (2024, 2025) draws the same conclusion based on patterns of floating coordinators. Weisser notes that some languages have coordinators that show an alternations between 2nd position and final position inside conjuncts. In Udihe (Tungusic; Russia), for instance, the coordinator *-de* is right attached to each conjunct in nominal coordination (60a), but appears in second position in clausal coordination (60b).

(60) *Alternation between 2nd and final position in Udihe:*

a. [$_{DP}$ Ami:-**de**] [$_{DP}$ eni:-**de**] [$_{DP}$ xa-nta-na-i-**de**]
father.2SG-FOC mother.2SG-sc foc sibling.N-PL-2SG-FOC
bi-si:-sini-gele-je.
be-IMPF-PERF.CVB-3SG-ask.IMP.2SG
'If you have a father, mother, or siblings, call (them).'

b. [$_{CP}$ Zuge-we-**de** čalu-wene-mi] [$_{CP}$ ima:-we-**de** čalu-wene-mi] [$_{CP}$ o:kto-**de**
ice-ACC-FOC melt-CAUS-1SG snow-ACC-FOC melt-CAUS-1SG grass-ACC-FOC
bagdi-wana-mi] …
grow-CAUS-1SG
'I melt the ice, I melt the snow, I make the grass grow …'
(Nikolaeva and Tolskaya 2001:548,648, cited in Weisser 2025)

Weisser argues that these patterns can be understood if such coordinators are underlyingly head-final, but may undergo an operation of lowering into each conjunct. This operation may target the 2nd position, since a specifier, although non-local in linear terms, is the highest phrase inside the conjunct and so may constitute the closest target. See Weisser (2024) for detailed argumentation.

Adamson (2022) reaches the same conclusion based on patterns in lowering of the definite article in Bulgarian. The Bulgarian definite article typically appears on the first morphological word in the noun phrase, so that a prenominal adjective must function as host (61a–b). But with adjectives that cannot carry inflection, many speakers allow the definite article to exceptionally skip over a modifier and be hosted by the noun (61c).

(61) *Bulgarian definite article can skip over inflectionless adjective:*

a. interesni-**jat** čovek
interesting.MS-DEF person

b. *interesen čovek-**at**
interesting.MS person-DEF
'the interesting person'
c. %erbap žena-**ta**
skillful woman-DEF
'the skillful woman'
(Adamson 2022:1)

Adamson argues that this pattern reflects an operation of Lowering of a D head that targets the closest possible host, rather than necessarily the head of its complement.

The idea that the locality conditions on lowering ought to be relaxed is supported also by other apparent cases of non-local cliticization that operate downward. In Tiwa, focus clitics with sentential scope attach either to the verb or auxiliary (62a–b), which Dawson (2017) argues is the result of lowering.

(62) *Non-local cliticization of focus clitic in Tiwa:*
a. Lí thái-do=**sê**.
go AUX-IMPF=FOC
'They are still going.'
b. Lí=**sê** thái-do.
go=FOC aux-IMPF
'They are still going.'
(Tiwa; Dawson 2017)

Another pattern of successive lowering seems to be found in varieties in German, when we examine the placement of the infinitival marker *zu*. Infinitival *zu* acts like a proclitic that typically attaches to the final auxiliary or verb in a verb cluster (Salzmann 2019; Schallert 2020), which Salzmann (2019) analyzes as a case of lowering from a clause-final position. As a result, the morphosyntax of *zu* is generally comparable to the default pattern found with Tigrinya complementizers. But Schallert (2020) points out that there are dialects in which *zu* seems to be able to lower farther leftward and downward, across the closest verb or auxiliary in a verb cluster (63a–b). Such cases may even give rise to *zu*-doubling (63c), implicating intermediate steps of lowering, as in the Tigrinya pattern.

(63) *Non-local cliticization of zu in German varieties:*
a. Er is lieber humplig ham glofa, als sich vo mir **z**=fahra lo.
he is rather limping home walked than REFL from me to=drive let
'He rather walked home limping than let himself be driven home by me.'
(Vorarlberg Alemmanic; Schallert 2020:52)
b. Mei Våta glap **z**=gwing kinn.
my father believes to=win can
'My father believes he is able to win.'
(Eastern Tyrol; Mayerthaler et al. 1995:55, cited in Schallert 2020:52)
c. Ich brauch merr deß net **zu** gefalle **zu** gelasse.
I need me.DAT that not to please to let
'I don't need to put up with that.'
(Frankfurt; Brückner 1988:3651, cited in Schallert 2020:52)

Pots (2017) describes a related set of facts in Dutch. In Dutch, many speakers permit infinitival *te*, a cognate of German *zu*, to displace leftward in a head-initial verb cluster. In (64), the base position of *te* is before the final verb *wachten* 'wait', since *zitten* 'sit' can select for a complement headed by *te*. Pots investigates variation across speakers and shows that *te* can move leftward and appear in front of any higher verb in the same cluster.

(64) *Leftward raising of te in Dutch verb clusters:*

Peter zal vanwege de nieuwe dienstregeling binnenkort nog langer op de trein [**<te>**
Peter will because.of the new schedule soon even longer on the train to
moeten **<te>** zitten **<te>** wachten].
must.INF to sit.INF to wait.INF
'Because of the new schedule, Peter will soon have to wait even longer for the train.'

Such cases can similarly be thought of as non-local cliticization, but with an operation of raising (Harizanov and Gribanova 2019). Finally, Bruening (2019) discusses patterns of inflection in Maliseet-Passamaquoddy, in which high functional morphemes must attach to the verb despite the presence of an intervening auxiliary. Such cases may involve non-local Lowering rightward.

I conclude that Lowering of a head does not necessarily target the head of its complement (Harizanov and Gribanova 2019; Adamson 2022; Weisser 2024, 2025). Instead, I follow Embick and Noyer (2001:589) and adopt the idea that Lowering targets the *closest* head. I provide a definition of Lowering in (65).

(65) **Lowering:**

If X and Y are syntactic heads, X c-commands Y, and Y is the closest head to X, replace $[Y^0]$ with $[Y^0{+}X^0]$.

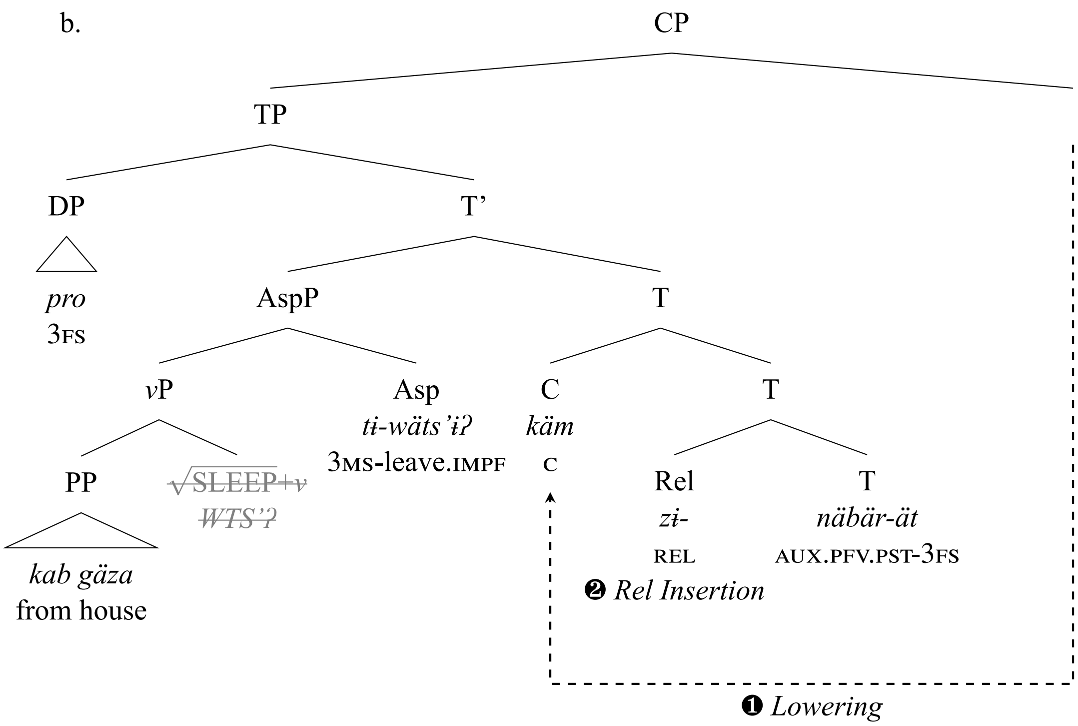
This view of Lowering defines the target of Lowering as the closest head. As I show below, this change will also allow for successive Lowering, if excorporation is permitted.

Defining the target of Lowering as the closest head allows for the cases discussed by Weisser (2024, 2025), in which a head-final conjunction particle targets the highest specifier instead of the head of the complement. In Tigrinya complementizer placement, though, the preferred target will still be the head of the complement of C, the head of the Tense Phrase. Consider the derivation of (66a) with Lowering and concomitant Rel Insertion again, in (66b).

(66) *Lowering in the postsyntactic derivation of (66a):*

a. **ካብ ገዛ ትወጽእ ክምዝነበረት ሓቢረ።**
[CP kab gäza tɨ-wäts'ɨʔ **käm**=zɨ-näbär-ät] ħabir-ä
from house 3FS-leave.IMPF C=REL-AUX.PFV.PST-3FS report-1S
'I reported that she was leaving the house.'

b.



Although the subject occupies the specifier of TP, Lowering in Tigrinya is motivated by a requirement of complementizers to precede a verbal element. As a result, the head of the complement is the closest target. The situation is different in the cases Weisser (2024, 2025) discusses, since many of the relevant conjunctive particles seem to prefer to attach to nominals. Thus, closeness for the purposes of Lowering must take into account whether a head is an *eligible* target for Lowering, a logic familiar from other minimality effects (Rizzi 1990). I provide a definition for closeness in (67).

(67) **Closeness (see also Weisser 2024:352):**
Y is the closest target for an operation initiated by X if there is no potential target Z that can satisfy the requirements of X such that X asymmetrically c-commands Z and Z asymmetrically c-commands Y.

This additional flexibility in the statement of Lowering allows for the coordinator alternations discussed by Weisser (2024, 2025) to be derived. In Tigrinya, this definition means that the closest target for Lowering of C in Tigrinya will be whatever verb or auxiliary occupies T. Although the subject in Spec-TP is c-commanded by C and itself c-commands the Tense head, it cannot satisfy the requirement of C to precede a verbal head. As a result, we continue to derive the preferred placement of complementizers before the clause-final verb or auxiliary.

The second proposal that is important to the account developed here is that I propose that excorporation is permitted in postsyntactic Lowering (see also Güneş 2021). I follow Kayne (1994:ch. 3.2) in adopting the idea that a head adjoined to another c-commands out of it. Kayne (1994) adopts a definition of c-command that restricts to c-command to *categories* (68), or all segments with the same label.

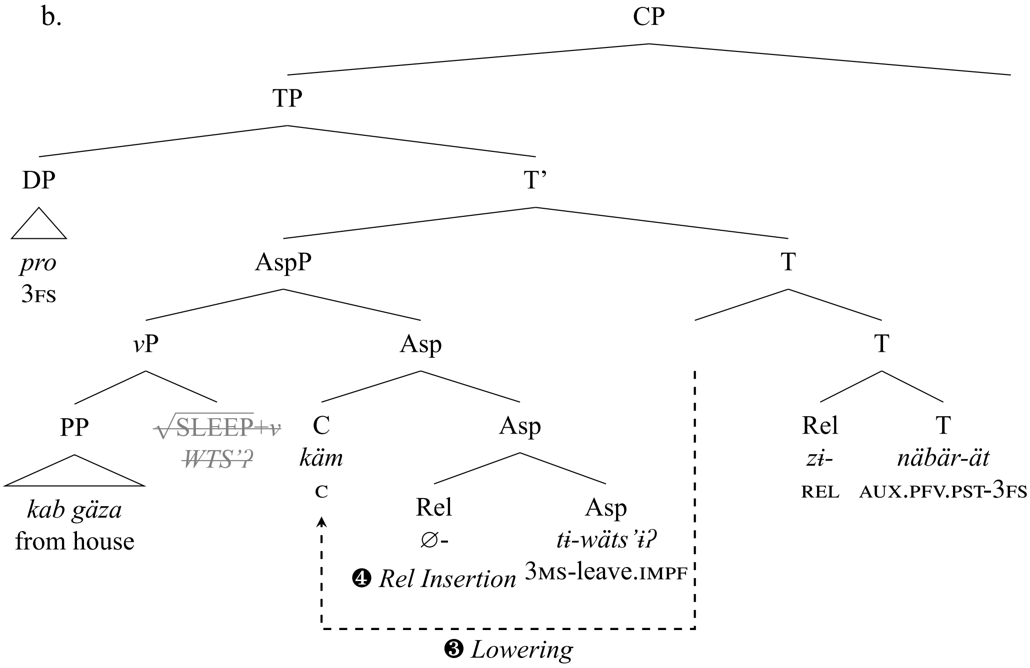
(68) **C-command (adapted from Kayne 1994:18):**
X c-commands Y iff X and Y are categories, X excludes Y, and every category that dominates X dominates Y.

In a complex head [$_T$ T C], the two segments with the label T act as one *category*. As a result, the category T as a whole does not dominate C, since only one of its segments does (the complex head). By the definition in (68), C ends up with the same c-command domain as T.[18] As a result, another step of Lowering is possible to deliver non-local cliticization in examples like (69a), with the complementizer excorporating and adjoining to the verb in Asp (69b).

(69) *Successive Lowering in postsyntactic derivation of (69a):*

a. **ካብ ገዛ ክምትወጽእ ዝነበረት ሓቢረ።**
[CP kab gäza **käm**=tɨ-wäts'ɨʔ zɨ-näbär-ät] ħabir-ä
from house C=3FS-leave.IMPF REL-AUX.PFV.PST-3FS report-1S
'I reported that she was leaving the house.'

b.



The starting point for this derivation is the structure created by the first application of Lowering and subsequent Rel Insertion in (66b). Another step of Lowering is licensed because C c-commands Asp by virtue of the definition of c-command in (68) and the fact that Lowering is not required to target the head of the complement. This step of Lowering is also followed by Rel Insertion, although the relative prefix is null before a prefixed base such as an imperfective. Subsequent to these postsyntactic operations, the structure undergoes bottom-up Vocabulary Insertion (Bobaljik 2000; Embick 2010). Because of the application of Rel Insertion in intermediate position, the effects of complementizer displacement remain visible on the clause-final auxiliary, which hosts relative morphology and surfaces with the perfective allomorph (as per the insertion rules in (11)).

In this derivation, the applications of Lowering and Rel Insertion are interleaved, with each step of Lowering obligatorily followed by node insertion. A question that arises is what rules out a derivation in which Lowering applies twice before Rel Insertion, which would result in the absence of intermediate morphological effects. I suggest that the ordering of morphological operations in instances of successive Lowering can be understood by thinking about their relationship to morphological wellformedness constraints. Both Lowering and Rel Insertion can be seen as responses to wellformedness conditions: Lowering satisfies the requirement of some complementizers to precede a verbal head, while Rel Insertion reflects a wellformedness condition on such configurations. I propose a general condition on the postsyntactic component that prioritizes operations that satisfy wellformedness conditions (70), based on Ringen's (1972) discussion of rule ordering in transformational grammar.

(70) **Obligatory Precedence Principle (Ringen 1972:270):**
Obligatory morphological operations apply before optional ones.

[18] In Kayne's (1994) definition, C c-commands T. But this view is not necessary for the current account.

This precedence principle generates the right ordering in Tigrinya complementizer displacement. The first step of Lowering is an obligatory response to a morphological wellformedness condition. This wellformedness condition is satisfied by adjunction to T, and an important consequence is that the second step of Lowering is not necessary to yield a grammatical output. In other words, successive Lowering is optional. As a result, the Obligatory Precedence Principle prioritizes the application of Rel Insertion in intermediate position, because Rel Insertion *does* satisfy a wellformedness condition, and guarantees that this operation must precede a second step of Lowering. Since Lowering feeds Rel Insertion, any instance of optional Lowering must still be followed by a step of Rel Insertion.

In this way, Tigrinya complementizer placement and its associated morphological effects arise through a morphosyntactic operation of lowering that can apply multiple times, as in Embick and Noyer's (2001) conception of Lowering.

Concluding remarks

This paper has presented a novel pattern of complementizer placement in the Ethio-Semitic language Tigrinya. Various complementizers can move leftward and downward through a head-final verb cluster, to attach to a verb or auxiliary, which I argued reflects an operation of Lowering in the sense of Embick and Noyer (2001). I showed that such displacement is associated with two morphological effects on the host of the complementizer: spreading of relative morphology and stem allomorphy. In cases of non-local cliticization, in which complementizers attach to a verb across an auxiliary, these morphological effects appear in intermediate positions, providing evidence for a conception of lowering as a morphosyntactic operation that can apply multiple times (Embick and Noyer 2001; Arregi and Pietraszko 2021).

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